

Mathematics Extended Essay

Modeling waves using the wave equation

How does the analytical solution to a wave propagating on a string fixed at both ends compare with and without the consideration of stretching in terms of waveform, period, instantaneous frequency and initial conditions?

Word count: 3991

Table of Contents

1. Introduction.....	4
1.1. Aim	4
1.2. Methodology.....	5
1.3. Background Theory	6
2. Solving the wave equation.....	12
2.1. Case 1: Without consideration of stretching.....	12
2.1.1 Derivation of the wave equation.....	12
2.1.2 Setting up the problem.....	16
2.1.3 Assumptions.....	17
2.1.4 Separation of variables.....	17
2.1.5 Applying boundary conditions.....	20
2.1.6 Superposition principle	21
2.1.7 Finding coefficients	21
2.2. Case 2: With consideration of stretching.....	24
2.2.1 Setting up the problem, Assumptions and Boundary conditions	24
2.2.2 Rederiving the wave equation.....	24
2.2.3 Solving the rederived wave equation (part 1).....	26
2.2.4 Prerequisites	27
2.2.5 Solving the rederived wave equation (part 2).....	35
2.2.6 Remarks on the solution.....	41
2.2.7 Finalising the solution.....	43
3. Comparison of the 2 cases	45
3.1. Physical parameters	45
3.2. Similiarity: Waveform	48
3.3. Difference 1: Period.....	49

3.4. Difference 2: Instantaneous frequency	56
3.5. Difference 3: Range of possible initial conditions.....	62
4. Conclusion	65
5. Appendix.....	67
6. Works Cited.....	77

1 Introduction:

1.1 Aim

This topic first became of interest when harmonic modes of standing waves was taught in IB physics classes. After research, it was realized that to quantitatively analyze the motion of propagating waves like standing waves, a partial differential equation (PDE) called the wave equation was required. Furthermore, it was found that the applications of such an equation extended to topics such as string theory, gravitational waves and electromagnetic waves (Unacademy). For example, the wave equation is critical in understanding electromagnetic waves and the fundamental properties of light such as its speed, interference patterns and polarization (Khan Academy). However, further research revealed that commonly used strings like guitars strings had high elastic modulus (a constant which determines the extent to which the length of the string is changed due to a change in tension (pulling force transmitted through a medium such as a string (Byjus)) (Instron)), causing the stretching of the string to have significant effects on the tension which are not accounted for in the original wave equation (Morse and Ingard). This inspired the research question: *How does the analytical solution to a wave propagating on a string fixed at both ends compare with and without the consideration of stretching in terms of waveform, period, instantaneous frequency and initial conditions?*

1.2 Methodology

The proceeding extended essay aims to use the wave equation to find and compare the analytical solution to the equation of motion of a string given an initial displacement of $f(x)$ and initial velocity of $g(x)$ with and without the consideration of stretching.

To find the solution without stretching, separation of variables is first employed (Herman). After which the superposition principle is applied as the final solution is a combination of various possible oscillation modes (Arias).

To take stretching into consideration, the original wave equation is modified to account for the change in tension caused by the elastic modulus and cross-sectional area and solved.

Finally, the waveform, period, instantaneous frequency and constraints on initial conditions of the analytical solution with and without the consideration of stretching is compared.

1.3 Background Theory

1.3.1 Linear Differential Equation

A linear differential equation in $y(x)$ is defined as a differential equation in the following form:

$$\sum_{i=0}^n p_i(x) \frac{d^i y}{dx^i} = g(x)$$

Where $p_i(x)$ and $g(x)$ are functions solely dependent on x (Boyce, DiPrima, and Meade).

1.3.2 Superposition principle

In general, the principle of superposition states that the overall result of 2 or more input variables simultaneously added is the same as the sum of each variable's respective individual result (Higa).

In the context of differential equations, the superposition principle states that for any linear differential equation with solutions $y_1(x)$, $y_2(x)$, $\dots$, $y_n(x)$

The general solution $y_g(t)$ is the linear combination of all solutions:

$$y_g(t) = \sum_{i=1}^n A_i y_i(x)$$

Where A_i is a constant (Zill and Cullen).

1.3.3 Partial Differential Equations

An ordinary differential equation (ODE) is an equation that involves ordinary derivatives (as opposed to partial derivatives) of a function (Nykamp). However, functions in a PDE are governed by more than 1 variable. For the case of the wave equation, they are position (x) and time (t).

For a function $y(x, t)$ its partial derivative can be taken with respect to x :

$$\frac{\partial y}{\partial x} = \lim_{h \rightarrow 0} \frac{y(x + h, t) - y(x, t)}{h}$$

Or it can be taken with respect to t :

$$\frac{\partial y}{\partial t} = \lim_{h \rightarrow 0} \frac{y(t + h, x) - y(t, x)}{h}$$

When taking the partial derivative with respect to a particular variable, all other variables (in this case just one other variable) are assumed to be constants and differentiated normally.

1.3.4 Force decomposition

Any force $\mathbf{F}$ can be decomposed into horizontal and vertical components by using trigonometry:

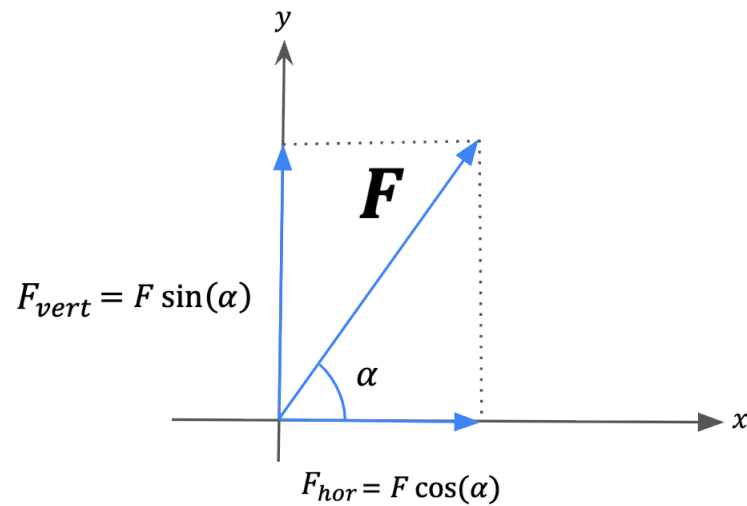


Diagram 1: force decomposition

1.3.5 Propagating waves

A wave is a disturbance that travels through a medium (Elert). The following diagram shows this phenomenon if the medium were a 1D string.

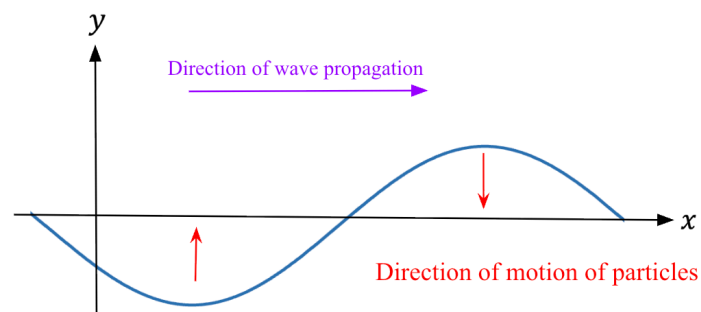


Diagram 2: A propagating wave

The general equation for a propagating wave on a 1D string is:

$$y(x, t) = A\sin(\omega t - kx) + B\cos(\omega t - kx) + C\sin(\omega t + kx) + D\cos(\omega t + kx) \quad (1)$$

Where A, B, C, D are constants (University of Texas).

1.3.6 Dispersion relation

For any propagating wave

$$V = \lambda f = \frac{\lambda}{P} \quad (2)$$

Where:

- V is the wave speed
- λ is the wavelength
- f is the frequency
- P is the period

When λ and f appear in trigonometric functions, they often appear as the wave number k and angular frequency ω .

Where:

$$\bullet \quad k = \frac{2\pi}{\lambda} \quad (3)$$

$$\bullet \quad \omega = 2\pi f \quad (4)$$

This is to adjust the period of the wave function to λ or P .

Consider $\sin(kx) = \sin\left(\frac{2\pi}{\lambda}x\right)$, k ensures that the function completes one full period

if x changes by λ , modifying sine's original period to match λ .

Now, consider $\sin(\omega t) = \sin(2\pi f t) = \sin\left(\frac{2\pi}{P}t\right)$, ω ensures that the function

completes one full period if t changes by P , modifying sine's original period to

match P .

1.3.7 Standing waves

Given specific initial conditions, a propagating wave can superpose with its reflected

wave to produce a wave that appears to be stationary, this is called a standing wave.

There are infinitely many ways this can occur, and each distinct pattern is called a

mode of oscillation.

The n^{th} oscillation mode is the n^{th} lowest frequency at which a standing wave is

formed.

For a string fixed at both ends, the following diagram shows its 1st, 2nd and 3rd oscillation mode:

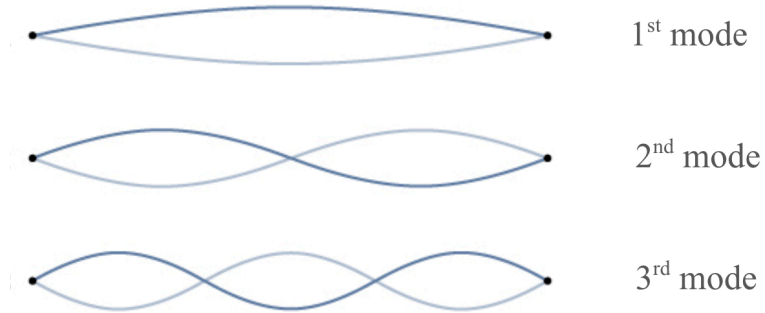


Diagram 3: first 3 oscillation modes for string fixed at both ends

(Pressbooks)

2. Solving the wave equation

2.1 Case 1: Without consideration of stretching

2.1.1 Derivation of the wave equation

The derivation will start by analyzing a small piece of a string on which a wave propagates. The string is located the point x_0 of the string, with a small length of Δx and has uniform linear mass density μ .

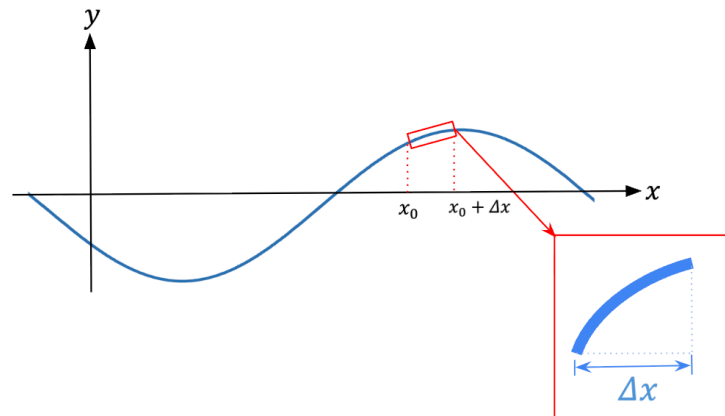


Diagram 4: focusing on a small part of the string

The tension force along the string is acting on both ends of the piece of string and can be decomposed into horizontal and vertical components.

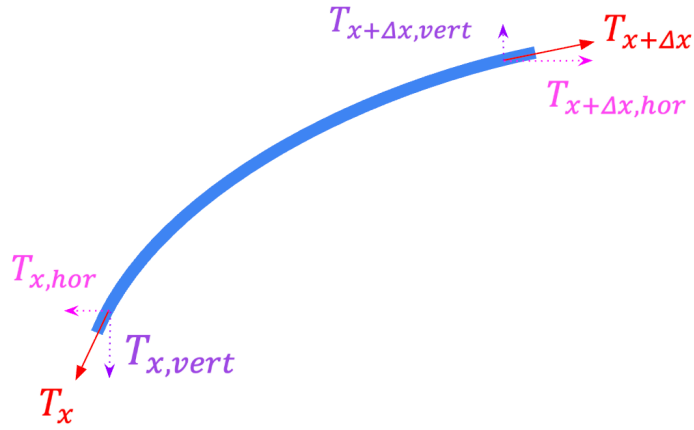


Diagram 5: force analysis on a small part of the string

This shows that the slope at x and $x + \Delta x$ can be expressed in terms of the ratio between the vertical and horizontal force.

$$\frac{T_{x,vert}}{T_{x,hor}} = \frac{\partial y}{\partial x} \Big|_{x=x_0} \quad (5)$$

$$\frac{T_{x+\Delta x,vert}}{T_{x+\Delta x,hor}} = \frac{\partial y}{\partial x} \Big|_{x=x_0+\Delta x} \quad (6)$$

Next, Newton's second law $F = ma$ (Jose, Jorge) is applied in the horizontal and vertical direction.

In the horizontal direction, there is no net force since during oscillations each part of the string simply moves up and down:

$$T_{x+\Delta x,hor} - T_{x,hor} = 0 \quad (7)$$

In the vertical direction:

$$T_{x+\Delta x, vert} - T_{x, vert} = \mu \Delta x a \quad (8)$$

where μ is the linear mass density.

Applying (5) and (6) to (8) yields:

$$T_{x+\Delta x, hor} \frac{\partial y}{\partial x} \Big|_{x=x_0+\Delta x} - T_{x, hor} \frac{\partial y}{\partial x} \Big|_{x=x_0} = \mu \Delta x a$$

Equation (7) implies that the $T_{x+\Delta x, hor} = T_{x, hor}$. For notational simplicity, a new variable T such that $T = T_{x+\Delta x, hor} = T_{x, hor}$ will be used,

This gives:

$$T \frac{\frac{\partial y}{\partial x} \Big|_{x=x_0+\Delta x} - \frac{\partial y}{\partial x} \Big|_{x=x_0}}{\Delta x} = \mu a$$

Note that the second term on the left hand side (LHS) of the equation is the second order partial derivative with respect to x if Δx approaches 0. Since Δx is small the expression can be approximated as:

$$T \frac{\partial^2 y}{\partial x^2} = \mu a$$

Since a , the acceleration, is the second order partial derivative of y with respect to time, this yields:

$$\frac{\partial^2 y}{\partial x^2} = \frac{\mu}{T} \frac{\partial^2 y}{\partial t^2} \quad (9)$$

Recall from (1), the general form for a propagating wave is:

$$y(x, t) = A\sin(\omega t - kx) + B\cos(\omega t - kx) + C\sin(\omega t + kx) + D\cos(\omega t + kx)$$

Substituting this back into (9) yields

$$-k^2 y(x, t) = -\frac{\mu}{T} \omega^2 y(x, t)$$

Simplifying the expression by using (2), (3) and (4) yields:

$$\frac{\mu}{T} = \left(\frac{1}{f\lambda}\right)^2 = \frac{1}{V^2} \quad (10)$$

Substituting (10) into (9) yields:

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{V^2} \frac{\partial^2 y}{\partial t^2} \quad (11)$$

2.1.2 Setting up the problem

A string that is fixed at both ends undergoes small oscillations. Find $y(x, t)$ of a string that is given an initial displacement of $f(x)$ and an initial velocity of $g(x)$ ignoring the effects of stretching.

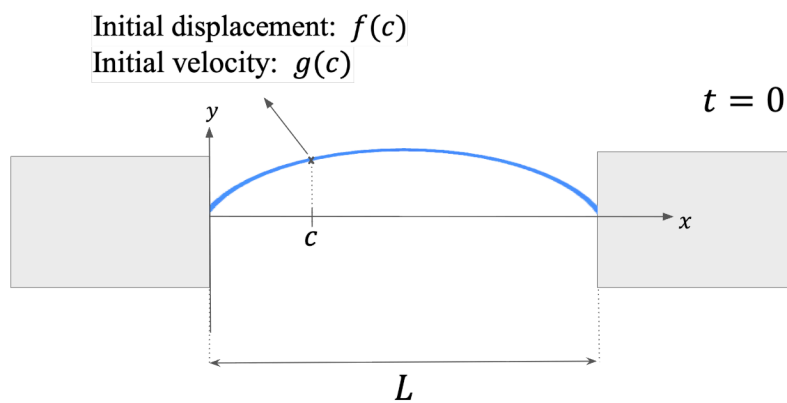


Diagram 6: initial conditions to string oscillation

As shown in Diagram 6, at time $t = 0$, any arbitrary point on the string with x coordinate $x = c$ will have a displacement of $f(c)$ and velocity of $g(c)$.

2.1.3 Assumptions

1. The stretching of the string is negligible
2. There is no damping
3. The linear mass density of the string is uniform
4. The amplitude of oscillations are small

2.1.4 Separation of Variables

It is assumed that the solution takes the form $X(x)T(t)$, where $X(x)$ is a function solely dependent on $T(t)$ is a function solely dependent on t (Herman).

Substituting into (11) yields:

$$\begin{aligned}\frac{\partial^2}{\partial x^2}(X(x)T(t)) &= \frac{1}{V^2} \frac{\partial^2}{\partial t^2}(X(x)T(t)) \\ X''(x)T(t) &= \frac{1}{V^2} T''(t)X(x) \\ \frac{X''(x)}{X(x)} &= \frac{1}{V^2} \frac{T''(t)}{T(t)}\end{aligned}\tag{12}$$

Because the LHS of (11) is an equation in terms of x whilst the right-hand side (RHS) is solely dependent on t , both sides must equal a constant. Let this constant be $-c^2$.

This yields:

$$\begin{cases} \frac{X''(x)}{X(x)} = -c^2 \\ \frac{1}{V^2} \frac{T''(x)}{T(x)} = -c^2 \end{cases}$$

Which yields:

$$\begin{cases} X''(x) + c^2 X(x) = 0 \\ T''(t) + (Vc)^2 T(t) = 0 \end{cases}$$

To solve $X''(x) + c^2 X(x) = 0$ note that the only type of function where it's

derivative is constant multiples of itself is e^x .

By ansatz (an educated guess on an unknown function), let $X(x) = e^{px}$, where p is a constant.

$$p^2 e^{px} + c^2 e^{px} = 0$$

$$e^{px}(p^2 + c^2) = 0$$

Since $e^{px} > 0$

$$p^2 + c^2 = 0$$

$$p^2 = -c^2$$

$$p = +ic \text{ or } -ic$$

Recognizing $X''(x) + c^2 X(x) = 0$ as a linear differential equation, the superposition principle for linear differential equations is applied. Therefore, the general solution to this differential equation is the linear combination of all possible solutions.

$$X(x) = Ae^{icx} + Be^{-icx}$$

Applying Euler's formula yields:

$$X(x) = A(\cos(cx) + i \sin(cx)) + B(\cos(-cx) - i \sin(-cx))$$

$$= A \cos(cx) + B \cos(cx) + Ai \sin(cx) + Bi \sin(cx)$$

Defining $A' = A + B$ and $B' = Ai + Bi$ yields the general form of $X(x)$.

$$X(x) = A' \cos(cx) + B' \sin(cx)$$

Applying the same analysis to $T(t)$ yields the following general form for the solutions of $X(x)$ and $T(t)$.

$$\begin{cases} X(x) = A \sin(cx) + B \cos(cx) \\ T(t) = C \sin(Vct) + D \cos(Vct) \end{cases}$$

Therefore, the final solution takes the form:

$$X(x)T(t) = (A \sin(cx) + B \cos(cx))(C \sin(Vct) + D \cos(Vct))$$

Using trigonometric identities, an alternate general form for $X(x)T(t)$ can be obtained:

$$X(x)T(t) = A \sin(Vct - cx) + B \cos(Vct - cx) + C \sin(Vct + cx) + D \cos(Vct + cx)$$

By comparing the above expression with the general form of a propagating wave:

$$y(x, t) = A \sin(\omega t - kx) + B \cos(\omega t - kx) + C \sin(\omega t + kx) + D \cos(\omega t + kx)$$

It can be seen that each term can be matched with a correspondent term in the general form, implying that the constant c is the wave number k .

2.1.5 Applying boundary conditions

Since the string is fixed at both ends, the boundary condition is:

$$y(0, t) = y(L, t) = 0$$

This is called the “Dirichlet boundary condition” (Williams) and yields:

$$y(0, t) = (B)(C \sin(Vkt) + D \cos(Vkt)) = 0$$

$$\Rightarrow B = 0$$

$$y(L, t) = (A \sin(kL))(C \sin(Vkt) + D \cos(Vkt)) = 0$$

$$\Rightarrow kL = n\pi, n \in \mathbb{N}$$

$$\Rightarrow k_n = \frac{n\pi}{L}$$

(Williams)

The value of each k_n is called an eigenvalue, a particular wave number for which a non-trivial solution exists that satisfies both the wave equation and the boundary conditions.

When k_n is substituted back, $\sin\left(\frac{n\pi}{L}x\right)$ is obtained. This is the eigenfunction, a function which represents a particular mode of oscillation, which corresponds to the n^{th} mode.

Thus, the solution for an oscillation of the n^{th} mode is:

$$y(x, t) = \left(A_n \sin\left(\frac{n\pi}{L}Vt\right) + B_n \cos\left(\frac{n\pi}{L}Vt\right)\right) \left(\sin\left(\frac{n\pi}{L}x\right)\right)$$

Note that the coefficient of $\sin\left(\frac{n\pi}{L}x\right)$ is absorbed into the coefficients A_n and B_n .

2.1.6 Superposition Principle

Applying the superposition principle, the full solution of $y(x, t)$ should be the sum of all eigenfunctions:

$$y(x, t) = \sum_{n=1}^{\infty} \left(A_n \sin\left(\frac{n\pi}{L} Vt\right) + B_n \cos\left(\frac{n\pi}{L} Vt\right) \right) \left(\sin\left(\frac{n\pi}{L} x\right) \right)$$

2.1.7 Finding coefficients

To find A_n and B_n , the orthogonal relations of trigonometric functions is used:

$$\int_0^L \sin\left(\frac{n\pi}{L} x\right) \sin\left(\frac{m\pi}{L} x\right) dx = \begin{cases} 0, & n \neq m \\ \frac{L}{2}, & n = m \end{cases}$$

(Kreyszig)

Proof:

If $n \neq m$:

$$\begin{aligned} \int_0^L \sin\left(\frac{n\pi}{L} x\right) \sin\left(\frac{m\pi}{L} x\right) dx &= \int_0^L \frac{1}{2} \left[\cos\left(\frac{(n-m)\pi}{L} x\right) - \cos\left(\frac{(n+m)\pi}{L} x\right) \right] dx \\ &= \frac{1}{2} \left[\left(\frac{L}{(n-m)\pi} \right) \sin\left(\frac{(n-m)\pi}{L} x\right) - \left(\frac{L}{(n+m)\pi} \right) \sin\left(\frac{(n+m)\pi}{L} x\right) \right]_0^L \\ &= \frac{1}{2} \left[\left(\frac{L}{(n-m)\pi} \right) \sin((n-m)\pi) - \left(\frac{L}{(n+m)\pi} \right) \sin((n+m)\pi) \right] \end{aligned}$$

Note that $\sin((n-m)\pi) = 0$ and $\sin((n+m)\pi) = 0$ for any $n, m \in \mathbb{N}$ and since

$n \neq m$, $\frac{L}{(n-m)\pi}$ is well-defined.

Therefore:

$$\int_0^L \sin\left(\frac{n\pi}{L} x\right) \sin\left(\frac{m\pi}{L} x\right) dx = \frac{1}{2} \left[\left(\frac{L}{(n-m)\pi} \right) (0) - \left(\frac{L}{(n+m)\pi} \right) (0) \right] = 0$$

If $n = m$:

$$\begin{aligned}
 \int_0^L \sin\left(\frac{n\pi}{L}x\right) \sin\left(\frac{m\pi}{L}x\right) dx &= \int_0^L \sin^2\left(\frac{n\pi}{L}x\right) dx \\
 &= \int_0^L \frac{1}{2} - \frac{1}{2} \cos\left(\frac{2n\pi}{L}x\right) dx \\
 &= \left[\frac{1}{2}x - \left(\frac{n\pi}{L}\right) \sin\left(\frac{2n\pi}{L}x\right) \right]_0^L \\
 &= \frac{L}{2}
 \end{aligned}$$

□

Considering $y(x, 0)$

$$y(x, 0) = f(x) = \sum_{n=1}^{\infty} (B_n) \left(\sin\left(\frac{n\pi}{L}x\right) \right)$$

Multiply both sides by $\sin\left(\frac{m\pi}{L}x\right)$, $m \in \mathbb{N}$:

$$\sin\left(\frac{m\pi}{L}x\right) f(x) = \sum_{n=1}^{\infty} (B_n) \left(\sin\left(\frac{m\pi}{L}x\right) \right) \left(\sin\left(\frac{n\pi}{L}x\right) \right)$$

Integrating both sides from 0 to L yields:

$$\int_0^L \sin\left(\frac{m\pi}{L}x\right) f(x) dx = \sum_{n=1}^{\infty} (B_n) \int_0^L \left(\sin\left(\frac{m\pi}{L}x\right) \right) \left(\sin\left(\frac{n\pi}{L}x\right) \right) dx$$

Applying the orthogonality property yields:

$$\int_0^L \sin\left(\frac{m\pi}{L}x\right) f(x) dx = B_m \left(\frac{L}{2} \right)$$

Therefore, the general formula for B_n is:

$$B_n = \left(\frac{2}{L} \right) \int_0^L \sin\left(\frac{n\pi}{L}x\right) f(x) dx$$

Now consider $\frac{\partial y}{\partial t}\Big|_{t=0}$

$$\frac{\partial y}{\partial t}\Big|_{t=0} = g(x) = \sum_{n=1}^{\infty} (A_n) \left(\frac{n\pi V}{L}\right) \left(\sin\left(\frac{n\pi}{L}x\right)\right)$$

Multiplying both sides by $\sin\left(\frac{m\pi}{L}x\right)$ and integrating both sides from 0 to L yields:

$$\int_0^L \left(\sin\left(\frac{m\pi}{L}x\right)\right) g(x) dx = \sum_{n=1}^{\infty} (A_n) \left(\frac{n\pi V}{L}\right) \int_0^L \left(\sin\left(\frac{m\pi}{L}x\right)\right) \left(\sin\left(\frac{n\pi}{L}x\right)\right) dx$$

Applying the orthogonality property yields:

$$\begin{aligned} \int_0^L \left(\sin\left(\frac{m\pi}{L}x\right)\right) g(x) dx &= (A_m) \left(\frac{m\pi V}{L}\right) \left(\frac{L}{2}\right) \\ A_n &= \frac{2}{n\pi V} \int_0^L \left(\sin\left(\frac{n\pi}{L}x\right)\right) g(x) dx \end{aligned}$$

2.2 With consideration of stretching

2.2.1 Problem setup, assumptions and boundary conditions

The same problem statement and setup as the first case is used but without neglecting the effects of stretching. Thus, the assumptions are same as case 1 except that the assumption of no stretching is eliminated. The same boundary conditions as case 1 are considered.

2.2.2 Rederiving the wave equation

The effect of the stretching can be accounted by considering Hooke's Law which states the effect that the change in the length of the string has on the tension which is assumed to be uniform.

$$T' = T + E a_c \frac{S - L}{L}$$

Where:

- E is the elastic modulus
- T is the tension of the string omitting the effects of stretching given by

$$T = V\mu \text{ (} V \text{ is the wave speed and } \mu \text{ is the string's linear mass density)}$$

- T' is the tension considering the effects of stretching
- a_c is the cross-sectional area of the string
- L is the original length of the string
- S is the length of the string after stretching (Oplinger)

S can be obtained by using the standard integral method of finding arc lengths:

$$S = \int_0^L \sqrt{1 + \left(\frac{\partial y}{\partial x}\right)^2} dx$$

Because the amplitude of oscillations is still assumed to be small, $\frac{\partial y}{\partial x}$ is assumed to be small, enabling the application of the binomial approximation to obtain the following expression:

$$\begin{aligned} S &\approx \int_0^L 1 + \frac{1}{2} \left(\frac{\partial y}{\partial x}\right)^2 dx \\ S &\approx L + \frac{1}{2} \int_0^L \left(\frac{\partial y}{\partial x}\right)^2 dx \end{aligned}$$

Therefore, T' can be expressed as:

$$T' = T + \frac{1}{2} \frac{E a_c}{L} \int_0^L \left(\frac{\partial y}{\partial x}\right)^2 dx \quad (13)$$

From (9), the wave equation for string with tension T' can be expressed as:

$$\frac{\partial^2 y}{\partial t^2} - \frac{T'}{\mu} \left(\frac{\partial^2 y}{\partial x^2}\right) = 0$$

Substituting (13) into the above expression yields:

$$\frac{\partial^2 y}{\partial t^2} - \frac{T}{\mu} \left(1 + \frac{E a_c}{2TL} \int_0^L \left(\frac{\partial y}{\partial x}\right)^2 dx\right) \left(\frac{\partial^2 y}{\partial x^2}\right) = 0$$

Substituting (10) for $\frac{T}{\mu}$ yields:

$$\frac{\partial^2 y}{\partial t^2} - V^2 \left(1 + \frac{E a_c}{2TL} \int_0^L \left(\frac{\partial y}{\partial x}\right)^2 dx\right) \left(\frac{\partial^2 y}{\partial x^2}\right) = 0$$

The above is the rederived wave equation when stretching is considered.

2.2.3 Solving the rederived wave equation (part 1)

Again, separation of variables is used.

Assuming $y(x, t) = T(t)X(x)$, $\int_0^L \left(\frac{\partial y}{\partial x}\right)^2 dx$ can be simplified.

$$\int_0^L \left(\frac{\partial y}{\partial x}\right)^2 dx = T(t)^2 \int_0^L \left(\frac{dX}{dx}\right)^2 dx$$

Substituting back into the wave equation yields:

$$\frac{\partial^2 y}{\partial t^2} - V^2 \left(1 + \frac{Ea_c}{2TL} T(t)^2 \int_0^L \left(\frac{dX}{dx}\right)^2 dx \right) \left(\frac{\partial^2 y}{\partial x^2}\right) = 0$$

Substituting $T(x)X(x)$ for all y and rearranging terms yields:

$$\frac{T''(t)}{V^2 \left(1 + \frac{Ea_c}{2TL} T(t)^2 \int_0^L \left(\frac{dX}{dx}\right)^2 dx \right) T(t)} = \frac{X''(x)}{X(x)}$$

Assuming these 2 expressions equate to $-k^2$ where k is a constant yields:

$$\begin{cases} T''(t) + k^2 V^2 \left(1 + \frac{Ea_c}{2TL} T(t)^2 \int_0^L \left(\frac{dX}{dx}\right)^2 dx \right) T(t) = 0 \\ X''(x) + k^2 X(x) = 0 \end{cases}$$

Where k is the wave number.

Ansatz cannot be used to solve the differential equation that yields the temporal part

of the solution. Instead, the Jacobean Elliptic Function is needed (Oplinger).

To apply this function, there are several prerequisites which will covered in the following section.

2.2.4 Prerequisites

2.2.4.1 Geometry of Ellipses

1. The ellipse

An ellipse is a collection of points such that the sum of the distances between each point and 2 other points called the foci is constant. (Byju's)

The general equation of an ellipse is:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Where a and b are constants.

In the derivation, ellipses with $b = 1$ are considered. Therefore, the equation of the ellipse becomes:

$$\frac{x^2}{a^2} + y^2 = 1 \quad (14)$$

This produces an ellipse with the following features:

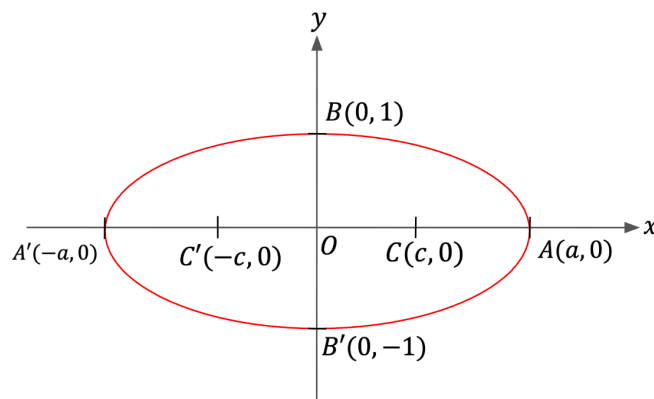


Diagram 7: graph of an ellipse

Where C and C' are the foci of the ellipse.

2. Eccentricity

The following sections introduces 2 formulae to calculate eccentricity for ellipses in the form of (14).

The eccentricity can be defined as such:

$$e = \frac{c}{a} \quad (15)$$

Intuitively, this describes how “squished” or “stretched” the ellipse is.

Note that $e \in (0, 1)$. The ellipse becomes a circle as e approaches 0 and becomes a horizontal line as e approaches 1.

Consider the distances between the foci and point A (That is, $C'A$ and CA):

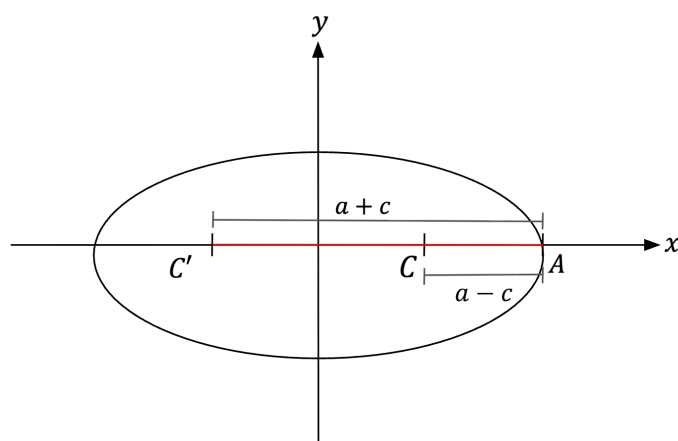


Diagram 8: $C'A$ and CA

Note that the sum of the 2 lines is $(a - c) + (a + c) = 2a$. From the definition of an ellipse, this implies that the sum of the distances between any point on the ellipse to each of the foci will always be $2a$.

Now, consider the isosceles triangle formed by the lines $C'B$, CB and $C'C$:

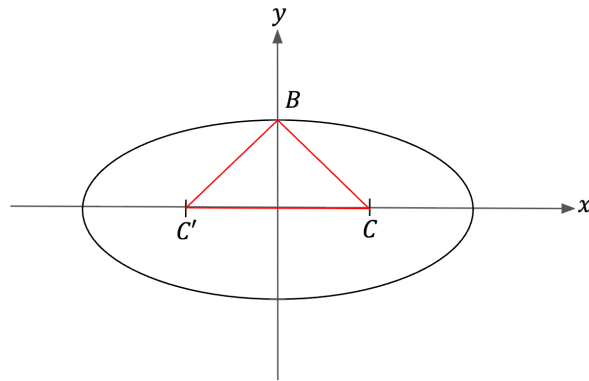


Diagram 9: triangle formed by $C'C$, $C'B$ and CB

By definition, $C'B + CB = 2a$. And since the triangle $C'CB$ is isosceles, this implies that $C'B = CB = a$.

Therefore, focusing on the right half of the triangle yields this triangle:

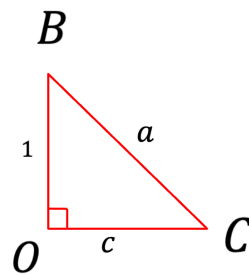


Diagram 10: right half of the triangle

By the Pythagorean theorem:

$$c = \sqrt{a^2 - 1^2} \quad (16)$$

Substituting (16) into (15) yields:

$$e = \sqrt{1 - \frac{1}{a^2}} \quad (17)$$

2.2.4.2 Jacobean elliptical functions

The following section aims to introduce and derive identities relating to Jacobean Elliptic Functions.

Analogous to how any point on a circle can be described with trigonometric functions, any point on the ellipse can be described using Jacobean elliptical functions.

Let $P(x, y)$ be an arbitrary point on an ellipse.

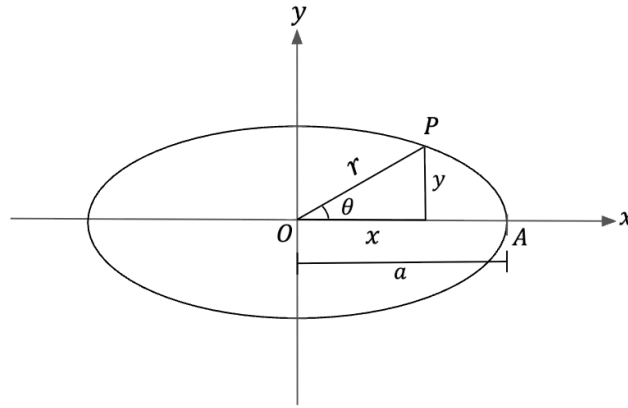


Diagram 11: point P on the ellipse

The Jacobean Elliptic Functions associated to point P are

$\text{sn}(u, e)$, $\text{cn}(u, e)$ and $\text{dn}(u, e)$. They are defined as follows:

$$\text{sn}(u, e) = y \tag{18}$$

$$\text{cn}(u, e) = \frac{x}{a} \tag{19}$$

$$\text{dn}(u, e) = \frac{r}{a} \tag{20}$$

Where:

- r is the distance between the center of the ellipse to P
- u is defined such that

$$du = rd\theta \quad (21)$$

(Kovacic, Cveticanin et al).

2.2.4.3 Elliptical trigonometric identities

The following section aims to prove the identities (22) and (23).

Substituting (18) and (19) into (14) yields:

$$\text{sn}^2(u, e) + \text{cn}^2(u, e) = 1 \quad (22)$$

By the Pythagorean theorem:

$$x^2 + y^2 = r^2$$

Dividing both sides by a^2 :

$$\left(\frac{x}{a}\right)^2 + \left(\frac{y}{a}\right)^2 = \left(\frac{r}{a}\right)^2$$

Substituting (18), (19) and (20) into the above expression yields:

$$a^2 \text{cn}^2(u, e) + \text{sn}^2(u, e) = a^2 \text{dn}^2(u, e)$$

Substituting (17) and (19) into the above expression yields:

$$\text{dn}^2(u, e) + e^2 \text{sn}^2(u, e) = 1 \quad (23)$$

2.2.4.4 Elliptical integral of the first kind

To derive an expression in which $\text{sn}(u, e)$ can be implicitly found as an integral, an alternate expression for du is utilized.

Recall from (21) that

$$du = r d\theta$$

The following substitution is utilized:

$$d\theta = \frac{(x^2 + a^2 y^2) dy}{x r^2}$$

Proof:

Consider

$$\tan \theta = \frac{y}{x}$$

Taking the derivative with respect to θ yields:

$$\sec^2(\theta) d\theta = \frac{x dy - y dx}{x^2}$$

Since

$$\sec(\theta) = \frac{r}{x}$$

This implies

$$d\theta = \frac{x dy - y dx}{r^2}$$

dx is expressed in terms of x, y, a and dy by considering the derivative with respect

to x of the ellipse equation

$$\frac{d}{dx} \left(\left(\frac{x}{a} \right)^2 + y^2 \right) = \frac{d}{dx} (1)$$

This implies:

$$dx = \frac{-a^2 y dy}{x}$$

Substituting back into the expression for $d\theta$ yields:

$$d\theta = \frac{(x^2 + a^2 y^2) dy}{x r^2}$$

□

Therefore:

$$\begin{aligned} du &= r d\theta \\ &= r \frac{(x^2 + a^2 y^2) dy}{x r^2} \\ &= \frac{(x^2 + a^2 y^2) dy}{x r} \end{aligned}$$

From (14), $x^2 + a^2 y^2 = a^2$. Thus,

$$du = \frac{a^2 dy}{x r}$$

Using this expression, an expression for the derivative of $\text{sn}(u, e)$ with respect to u

can be derived:

$$\frac{d(\text{sn}(u, e))}{du} = \left(\frac{xr}{a^2}\right) \left(\frac{d(\text{sn}(u, e))}{dy}\right)$$

$\text{sn}(u, e) = y \Rightarrow \frac{d(\text{sn}(u, e))}{dy} = 1$, therefore:

$$\frac{d(\text{sn}(u, e))}{du} = \left(\frac{xr}{a^2}\right)$$

Substituting (19) and (20) yields:

$$\frac{d(\text{sn}(u, e))}{du} = (\text{cn}(u, e))(\text{dn}(u, e)) \quad (24)$$

Next, rearranging the above equation yields:

$$du = \frac{d(\text{sn}(u, e))}{(\text{cn}(u, e))(\text{dn}(u, e))}$$

Substituting (22) and (23) into the above expression yields:

$$du = \frac{d(\text{sn}(u, e))}{(\sqrt{1 - (\text{sn}(u, e))^2})(\sqrt{1 - e^2(\text{sn}(u, e))^2})}$$

Integrating the LHS from $u = 0$ to some arbitrary u results in the bounds of

integration on the RHS to be from $\text{sn}(0, e) = 0$ to $\text{sn}(u, e)$:

$$\int_0^u du = u = \int_0^{\text{sn}(u, e)} \frac{dy}{(\sqrt{1 - y^2})(\sqrt{1 - e^2 y^2})}$$

Note that $\text{sn}(u, e)$ is substituted by y as it has become a dummy variable of the integrand.

Let $y = \sin\theta$

$$u = \int_0^{\arcsin(\text{sn}(u, e))} \frac{d\theta}{\sqrt{1 - e^2 \sin^2(\theta)}} \quad (25)$$

Therefore, for an integral in the form of $\int_0^t \frac{d\theta}{\sqrt{1 - e^2 \sin^2(\theta)}}$ where e is a constant, t will

be such that:

$$t = \arcsin(\text{sn}(u, e)) \Rightarrow \sin(t) = \text{sn}(u, e)$$

where u is the value of the integral $\int_0^t \frac{d\theta}{\sqrt{1 - e^2 \sin^2(\theta)}}$ (Wolfram).

2.2.5 Solving the rederived wave equation (part 2)

The differential equation that gives the temporal part of the solution will now be solved by adjusting it to the same form as (25), allowing $T(t)$ to be expressed as a Jacobean Elliptic Function.

$$T''(t) + k^2 V^2 \left(1 + \frac{E a_c}{2TL} T(t)^2 \int_0^L \left(\frac{dX}{dx} \right)^2 dx \right) T(t) = 0 \quad (26)$$

Let

$$a = k^2 V^2$$

$$b = \frac{k^2 V^2 E a_c}{2TL} \int_0^L \left(\frac{dX}{dx} \right)^2 dx$$

Note that since all physical constants (k, V, E, a_c, T, L) are positive and the integral

$\int_0^L \left(\frac{dX}{dx} \right)^2 dx$ is positive, it follows that $a > 0, b > 0$.

After the substitution, (26) becomes:

$$T''(t) + aT(t) + bT(t)^3 = 0$$

Multiplying both sides by $T'(t)$ yields:

$$T'(t)T''(t) + aT(t)T'(t) + bT(t)^3T'(t) = 0$$

$$\frac{d}{dt} \left[\frac{1}{2} T'(t)^2 + \frac{a}{2} T(t)^2 + \frac{b}{4} T(t)^4 \right] = 0$$

Integrating both sides and applying the fundamental theorem of calculus yields:

$$\frac{1}{2} T'(t)^2 + \frac{a}{2} T(t)^2 + \frac{b}{4} T(t)^4 = C$$

Where C is a constant.

Let the amplitude of $T(t)$ occur at t_{max} and let $T(t_{max}) = A$. Since this is a maximum point of $T(t)$, $T'(t_{max}) = 0$.

At $t = t_{max}$:

$$\frac{a}{2}A^2 + \frac{b}{4}A^4 = C$$

At $t = 0$:

$$\frac{1}{2}T'(0)^2 + \frac{a}{2}T(0)^2 + \frac{b}{4}T(0)^4 = C$$

Equating the above 2 equations yields:

$$A^2 + \frac{b}{4}A^4 = \frac{1}{2}T'(0)^2 + \frac{a}{2}T(0)^2 + \frac{b}{4}T(0)^4 \quad (27)$$

Using the quadratic formula, A to be found from initial conditions.

A dimensionless variable $u(t)$ is introduced to normalize $T(t)$.

Let $T(t) = Au(t)$

$$\frac{1}{2}A^2u'(t)^2 + \frac{a}{2}A^2u(t)^2 + \frac{b}{4}A^4u(t)^4 = \frac{a}{2}A^2 + \frac{b}{4}A^4$$

Multiplying by $\frac{2}{A^2}$:

$$u'(t)^2 + au(t)^2 + \frac{b}{2}A^2u(t)^4 = a + \frac{b}{2}A^2$$

$$u'(t)^2 = a(1 - u(t)^2) + \frac{b}{2}A^2(1 - u(t)^4)$$

$$u'(t)^2 = (1 - u(t)^2)(a + \frac{b}{2}A^2(1 + u(t)^2))$$

Let $u(t) = \sin(\theta(t))$

$$[\theta'(t)\cos(\theta(t))]^2 = [1 - \sin^2(\theta(t))] \left[a + \frac{b}{2}A^2(1 + \sin^2(\theta(t))) \right]$$

$$\theta'(t)^2 = a + \frac{b}{2}A^2(1 + \sin^2(\theta(t)))$$

$$\theta'(t)^2 = a + \frac{b}{2}A^2 + \frac{b}{2}A^2\sin^2(\theta(t))$$

$$\frac{d\theta(t)}{dt} = \pm \sqrt{a + \frac{b}{2}A^2 + \frac{b}{2}A^2\sin^2(\theta(t))}$$

The sign here will be chosen to be the same as the sign of $T'(0)$ because $\theta'(0)$ and

$T'(0)$ are linearly related.

Derivation :

From previous substitutions we have:

$$T(t) = Au(t) = A\sin(\theta(t)) \tag{28}$$

Differentiating with respect to time and substituting $t = 0$ yields:

$$T'(0) = A \cos(\theta(0)) \theta'(0)$$

Substituting $t = 0$ into (28) yields:

$$\sin(\theta(0)) = \left(\frac{T(0)}{A} \right)^2 \Rightarrow \cos(\theta(0)) = \sqrt{1 - \left(\frac{T(0)}{A} \right)^2}$$

Therefore:

$$T'(0) = \left(A \sqrt{1 - \left(\frac{T(0)}{A} \right)^2} \right) \theta'(0)$$

Now consider:

$$\frac{a}{2}A^2 + \frac{b}{4}A^4 = \frac{1}{2}T'(0)^2 + \frac{a}{2}T(0)^2 + \frac{b}{4}T(0)^4$$

$$a(A^2 - T(0)^2) + \frac{b}{2}(A^4 - T(0)^4) = T'(0)^2$$

$$(A^2 - T(0)^2)(a + \frac{b}{2}(A^2 + T(0)^2)) = T'(0)^2$$

Because the RHS is a square and thus positive, the LHS must also be positive.

Note that $(a + \frac{b}{2}(A^2 + T(0)^2))$ must be positive as both a and b are positive.

This implies:

$$A^2 - T(0)^2 \geq 0$$

$$A^2 \geq T(0)^2$$

$$A \geq |T(0)|$$

$$1 \geq \frac{|T(0)|}{A}$$

$$\Rightarrow A \sqrt{1 - \left(\frac{T(0)}{A}\right)^2} \geq 0$$

$$\Rightarrow T'(0) = k\theta'(0), k \in \mathbb{R}^+$$

Thus, $T'(0)$ and $\theta'(0)$ are linearly related.

Continuing from $\theta'(t)$:

$$\frac{d\theta(t)}{dt} = \pm \sqrt{a + \frac{b}{2}A^2 + \frac{b}{2}A^2 \sin^2(\theta(t))}$$

Let $\varphi(t) = \frac{\pi}{2} - \theta(t)$ which implies $\sin(\theta(t)) = \cos(\varphi(t))$ and $\frac{d\varphi(t)}{dt} = -\frac{d\theta(t)}{dt}$

$$\begin{aligned}\frac{d\varphi(t)}{dt} &= \mp \sqrt{a + \frac{b}{2}A^2 + \frac{b}{2}A^2\cos^2(\varphi(t))} \\ &= \mp \sqrt{a + \frac{b}{2}A^2 + \frac{b}{2}A^2(1 - \sin^2(\varphi(t)))} \\ &= \mp \sqrt{a + bA^2 - \frac{b}{2}A^2\sin^2(\varphi(t))} \\ &= \mp \sqrt{(a + bA^2) \left(1 - \frac{\frac{b}{2}A^2}{a + bA^2} \sin^2(\varphi(t))\right)}\end{aligned}$$

$$\text{Let } e^2 = \frac{\frac{b}{2}A^2}{a + bA^2} \quad (29)$$

Then, take the integral from $t = 0$ to some arbitrary t and denote the dummy variable in the integrand $\varphi(t)$ as φ :

$$\begin{aligned}\int_{\varphi(0)}^{\varphi(t)} \frac{d\varphi}{\sqrt{1 - e^2\sin^2(\varphi)}} &= \int_0^t \mp \sqrt{a + bA^2} dt \\ \int_0^{\varphi(t)} \frac{d\varphi}{\sqrt{1 - e^2\sin^2(\varphi)}} &= \int_0^t \mp \sqrt{a + bA^2} dt + \int_0^{\varphi(0)} \frac{d\varphi}{\sqrt{1 - e^2\sin^2(\varphi)}}\end{aligned}$$

This is in the same form as (25), enabling the application of elliptical trigonometric functions as previously derived

$$\sin(\varphi(t)) = \text{sn}\left(\int_0^t \mp \sqrt{a + bA^2} dt + \int_0^{\varphi(0)} \frac{d\varphi}{\sqrt{1 - e^2\sin^2(\varphi)}}, e\right)$$

By the Pythagorean theorem,

$$\cos(\varphi(t)) = \sqrt{1 - \sin^2(\varphi(t))}$$

Substituting $\sin(\varphi(t))$ using (19) yields:

$$\cos(\varphi(t)) = \sqrt{1 - sn^2 \left(\int_0^t \mp \sqrt{a + bA^2} dt + \int_0^{\varphi(0)} \frac{d\varphi}{\sqrt{1 - e^2 \sin^2(\varphi)}}, e \right)}$$

since $\sin(\theta(t)) = \cos(\varphi(t))$:

$$\sin(\theta(t)) = \text{cn} \left(\pm \sqrt{a + bA^2} t + \int_0^{\varphi(0)} \frac{d\varphi}{\sqrt{1 - e^2 \sin^2(\varphi)}}, e \right) \quad (30)$$

Note that shifting from $\varphi(t)$ back to $\theta(t)$ reverses sign of $\mp \sqrt{a + bA^2} t$.

Substituting (30) into $u(t) = \sin(\theta(t))$ yields:

$$u(t) = \text{cn} \left(\pm \sqrt{a + bA^2} t + \int_0^{\varphi(0)} \frac{d\varphi}{\sqrt{1 - e^2 \sin^2(\varphi)}}, e \right)$$

Next, undo the substitution $Au(t) = T(t)$:

$$T(t) = A \text{cn} \left(\pm \sqrt{a + bA^2} t + \int_0^{\varphi(0)} \frac{d\varphi}{\sqrt{1 - e^2 \sin^2(\varphi)}}, e \right)$$

Combining with the spatial solution under Dirichlet boundary condition yields:

$$y(x, t) = A \text{cn} \left(\pm \sqrt{a + bA^2} t + \int_0^{\varphi(0)} \frac{d\varphi}{\sqrt{1 - e^2 \sin^2(\varphi)}}, e \right) \left(\sin \left(\frac{n\pi}{L} x \right) \right)$$

Where n is the mode of oscillation.

The substitution $\omega = \sqrt{a + bA^2}$ and $\Phi = \int_0^{\varphi(0)} \frac{d\varphi}{\sqrt{1 - e^2 \sin^2(\varphi)}}$ yields the general form:

$$y(x, t) = A \text{cn}(\pm \omega t + \Phi, e) \left(\sin \left(\frac{n\pi}{L} x \right) \right)$$

This is similar to Oplinger's result:

$$y(x, t) = B \frac{\sin vx}{\sin vL} \operatorname{cn}(ft, K)$$

suggesting that the analytical solution is valid. The differences between the 2

solutions are due to the fact that Oplinger had assumed 0 initial velocity.

(Oplinger)

2.2.6 Remarks on the solution

2.2.6.1 Remark 1

It was assumed that there are real roots to (27).

This can be proven by considering the discriminant:

$$0 = \frac{b}{4}A^4 + \frac{a}{2}A^2 - \left(\frac{1}{2}T'(0)^2 + \frac{a}{2}T(0)^2 + \frac{b}{4}T(0)^4\right)$$

$$\Delta = \left(\frac{a}{2}\right)^2 + 4\left(\frac{b}{4}\right)\left(\frac{1}{2}T'(0)^2 + \frac{a}{2}T(0)^2 + \frac{b}{4}T(0)^4\right)$$

$b > 0$ implies that $4\left(\frac{b}{4}\right)\left(\frac{1}{2}T'(0)^2 + \frac{a}{2}T(0)^2 + \frac{b}{4}T(0)^4\right) > 0$ and since $\left(\frac{a}{2}\right)^2 > 0$,

$\Delta > 0$.

2.2.6.2 Remark 2

Unlike in case 1, the solution does not consist of infinitely many terms each corresponding to a mode of oscillation. This is because of the assumption that the solution consists only of one mode was made in the step:

$$\int_0^L \left(\frac{\partial y}{\partial x} \right)^2 dx = T(t)^2 \int_0^L \left(\frac{dX}{dx} \right)^2 dx$$

Assuming that the initial conditions consisted of 2 modes $T_1(t)X_1(x) + T_2(t)X_2(x)$:

$$\int_0^L \left(\frac{\partial y}{\partial x} \right)^2 dx = \int_0^L (T_1(t)X'_1(x))^2 + 2(T_1(t)X'_1(x))(T_2(t)X'_2(x)) + (T_2(t)X'_2(x))^2 dx$$

The additional term $2(T_1(t)X'_1(x))(T_2(t)X'_2(x))$ makes it impossible to factor out the temporal parts of the solution.

Thus, a limitation of the solution is that it only works if only one mode is excited by initial conditions.

This constrains the initial conditions to the analytical solution to one of the following

3 cases:

$$\begin{cases} f(x) = C \sin\left(\frac{N\pi}{L}x\right) \\ g(x) = D \sin\left(\frac{N\pi}{L}x\right) \end{cases}$$

$$\begin{cases} f(x) = 0 \\ g(x) = D \sin\left(\frac{N\pi}{L}x\right) \end{cases}$$

$$\begin{cases} f(x) = C \sin\left(\frac{N\pi}{L}x\right) \\ g(x) = 0 \end{cases}$$

where C and D are both constants and $N \in \mathbb{Z}^+$. Note that the case in which both

$f(x)$ and $g(x)$ are 0 is omitted due to it resulting in a trivial solution of

$$y(x, t) = 0.$$

2.2.7 Finalizing the solution

In any of the 3 cases, the N^{th} mode is excited, implying that $n = N$ in

$y(x, t) = A \cos(\pm \omega t + \Phi, e) \left(\sin \left(\frac{n\pi}{L} x \right) \right)$. Thus, the solution is given as follows:

$$y(x, t) = A \cos(\pm \omega t + \Phi, e) \left(\sin \left(\frac{N\pi}{L} x \right) \right)$$

To find the sign of the coefficient $\pm \sqrt{a + bA^2}$ and determine the value of A ,

e, φ_0 and Φ , $T(0)$ and $T'(0)$ must be found.

Assuming the initial conditions are of the following form

$$\begin{cases} f(x) = C \sin \left(\frac{N\pi}{L} x \right) \\ g(x) = D \sin \left(\frac{N\pi}{L} x \right) \end{cases}$$

Considering the initial displacement yields $T(0)$:

$$y(x, 0) = C \sin \left(\frac{N\pi}{L} x \right) = T(0) \sin \left(\frac{N\pi}{L} x \right)$$

$$C = T(0)$$

Considering the initial displacement gives $T'(0)$:

$$\frac{\partial y(x, 0)}{\partial t} = D \sin \left(\frac{N\pi}{L} x \right) = T'(0) \sin \left(\frac{N\pi}{L} x \right)$$

$$D = T'(0)$$

The sign of $\pm\sqrt{a + bA^2}$ will be chosen to be positive if $D \geq 0$ and vice versa if

$D < 0$. A similar procedure is utilised for the other 2 cases of initial conditions.

3. Comparison of the 2 cases

3.1 Physical parameters

To facilitate comparison between the analytical solution to case 1 and case 2, numerical values that simulate a guitar will be substituted into the solutions.

These physical parameters are taken:

Name	Symbol	Value
Tension (N)	T	72.2
Linear mass density (kg/m)	μ	7.25×10^{-4}
Length of String (m)	L	0.648
Elastic Modulus (Pa)	E	2.20×10^{11}
Cross Sectional Area (m^2)	a_c	9.24×10^{-8}

Table 1: Physical parameters of a “e-guitar steel B-string.” (Fletcher)

This results in these values:

Name	Symbol	Value
Wave speed(m/s)	V	316
Amplitude(m)	A	1
Eccentricity	e	0.701
N/A	a	2.34×10^6
N/A	b	2.80×10^{10}
N/A	φ_0	0
N/A	Φ	0

Table 2: Values for parameters in the analytical solutions

Note that all values that cannot be expressed as an integer is rounded to 3 significant figures for consistency.

The following shows the initial conditions which were chosen for monomodal oscillation:

$$\begin{cases} f(x) = \sin\left(\frac{\pi x}{L}\right) \\ g(x) = 0 \end{cases}$$

This yields the following solutions:

Case 1 (which will now be denoted as the linear wave):

$$y(x, t) = \cos(1530t)\sin(4.85x)$$

Derivation:

Recall that:

$$y(x, t) = \sum_n^{\infty} \left(A_n \sin\left(\frac{n\pi}{L} Vt\right) + B_n \cos\left(\frac{n\pi}{L} Vt\right) \right) \left(\sin\left(\frac{n\pi}{L} x\right) \right)$$

Substituting $t = 0$, $V = 316$ and $L = 0.648$ yields:

$$y(x, 0) = \sum_{n=1}^{\infty} B_n \sin(4.85nx) = \sin(4.85x)$$

$$\frac{\partial y}{\partial t} = - \sum_{n=1}^{\infty} 1530nA_n \sin(4.85nx) = 0$$

For the above to be true, all B_n except B_1 must be 0, all A_n must be 0 and $B_1 = 1$.

This implies:

$$y(x, t) = \cos(1530t)\sin(4.85x)$$

Since the initial conditions only excite one wave mode, the orthogonality property is

not needed to find the coefficients of the trigonometric functions.

Case 2 (which will now be denoted as the nonlinear wave):

$$\begin{aligned} y(x, t) &= A \operatorname{cn}(\omega t + \Phi, e) \sin\left(\frac{n\pi}{L} x\right) \\ &= A \operatorname{cn}\left(\sqrt{a + bA^2}t + \int_0^{\varphi(0)} \frac{d\varphi}{\sqrt{1 - e^2 \sin^2(\varphi)}}, e\right) \sin\left(\frac{n\pi}{L} x\right) \\ &= \operatorname{cn}(62200t, 0.701) \sin(4.85x) \end{aligned}$$

3.2 Similarity: Waveform

The following is an animation of the linear (stretching not considered) and nonlinear (stretching considered) wave:

https://drive.google.com/file/d/1Ft9EyJ5W_aXwn95lXz6Kf0JvuB4yTNzE/view?usp=drive_link

The following is a snapshot of the above animation taken at an arbitrary time:

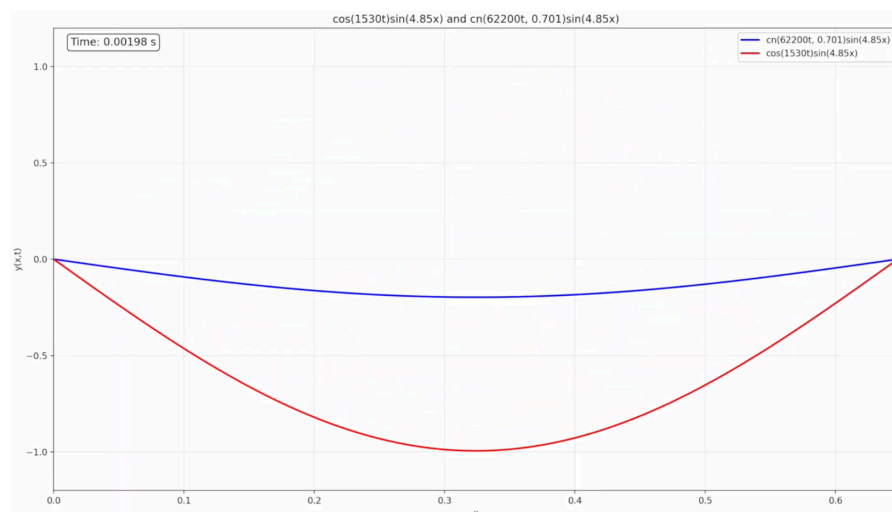


Diagram 12: linear and nonlinear wave at an arbitrary time

(“Give python code that animate the graphs of $\cos(1530t)\sin(4.85x)$ and $cn(62200t, 0.701)\sin(4.85x)$ from $t=0$ to $t=0.005$ ”) (Appendix A)

As seen from the Diagram 12, at any t , both analytical solutions’ waveforms are of a single sinusoidal arch.

This is because both solutions are in the form $y(x, t) = A(t)\sin(4.85x)$, where $A(t) \in [-1, +1]$. Therefore, at any time, both solutions are the same function only differing in the amplitude.

3.2 Difference 1: Period

From the previous animation, it can be observed that the period of the linear wave is larger than the nonlinear wave.

3.3.1 Qualitative explanation

By looking at PDE

$$\frac{\partial^2 y}{\partial t^2} - V^2 \left(1 + \frac{E a_c}{2TL} \int_0^L \left(\frac{\partial y}{\partial x} \right)^2 dx \right) \left(\frac{\partial^2 y}{\partial x^2} \right) = 0$$

For any physical parameters, $\frac{E a_c}{2TL} \int_0^L \left(\frac{\partial y}{\partial x} \right)^2 dx$ must be positive. Thus, the effective wave speed is higher in the nonlinear wave compared to the linear, leading the period to decrease.

3.3.2 Quantitative explanation

The period of the linear wave (denoted P_1) is:

$$P_1 = \frac{2\pi}{(1530)} \approx 4.11 \times 10^{-3}$$

The period of the nonlinear wave (denoted P_2) is:

$$P_2 = \frac{4(K(e) - \Phi)}{\omega} \approx 1.19 \times 10^{-4} \quad (31)$$

where $K(e) = \int_0^1 \frac{dy}{(\sqrt{1-y^2})(\sqrt{1-e^2y^2})}$ (Whittaker and Watson).

Derivation:

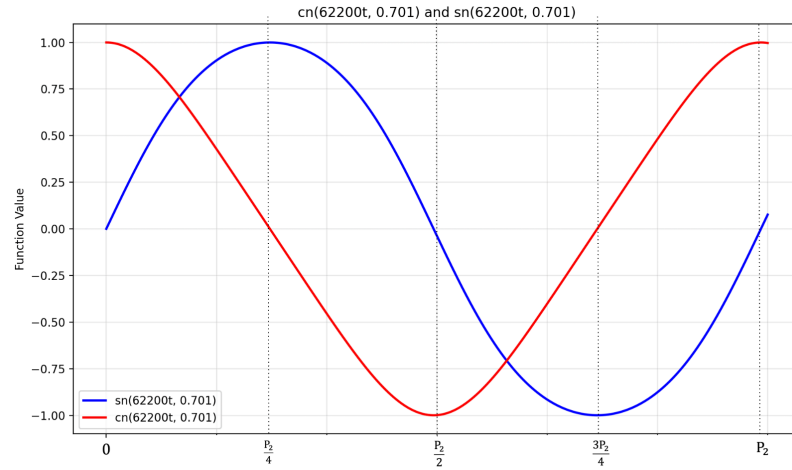


Diagram 13: one period of elliptical cosine and elliptical sine

(“Give python code that graphs $\text{cn}(62200t, 0.701)$ and $\text{sn}(62200t, 0.701)$ for one period”) (Appendix B)

Note that like sine and cosine, the periods of $\text{cn}(\omega t, e)$ and $\text{sn}(\omega t, e)$ are the same.

Furthermore, each period is divided into 4 parts in which the value of the function increases or decreases by 1.

Recall that:

$$u = \int_0^{\text{sn}(u,e)} \frac{dy}{(\sqrt{1-y^2})(\sqrt{1-e^2y^2})}$$

Let $u = \text{sn}(\omega t + \Phi, e)$

$$\omega t + \Phi = \int_0^{\text{sn}(\omega t + \Phi, e)} \frac{dy}{(\sqrt{1-y^2})(\sqrt{1-e^2y^2})}$$

From Diagram 12, notice that when $\text{sn}(\omega t + \Phi, e) = 1$ for the first time, $t = \frac{P_2}{4}$,

where P_2 is the period of $\text{sn}(\omega t + \Phi, e)$ and $\text{cn}(\omega t + \Phi, e)$.

$$\begin{aligned} \omega \left(\frac{P_2}{4} \right) + \Phi &= \int_0^1 \frac{dy}{(\sqrt{1-y^2})(\sqrt{1-e^2y^2})} \\ P_2 &= \frac{4 \left(\int_0^1 \frac{dy}{(\sqrt{1-y^2})(\sqrt{1-e^2y^2})} - \Phi \right)}{\omega} \\ &= \frac{4(K(e) - \Phi)}{\omega} \end{aligned}$$

Note that in standard notation, $\int_0^1 \frac{dy}{(\sqrt{1-y^2})(\sqrt{1-e^2y^2})}$ is denoted as $K(e)$ (Olver, Lozier et al).

Substituting $\omega = 62200$, $e = 0.701$ and $\Phi = 0$ yields:

$$\begin{aligned} P_2 &= \frac{4 \int_0^1 \frac{dy}{(\sqrt{1-y^2})(\sqrt{1-0.701^2y^2})}}{62200} \\ &\approx 1.19 \times 10^{-4} < 4.11 \times 10^{-3} = P_1 \end{aligned}$$

This quantitatively demonstrates why the linear wave has a larger period compared to the nonlinear wave.

3.3.3 Limiting Behaviour

1. When b approaches infinity

As the nonlinear term $b = \frac{Ea_c}{2TL} T(t)^2 \int_0^L \left(\frac{dx}{dx} \right)^2 dx$ approaches infinity due to extreme

physical constraints (For example if E approaches infinity), P_2 should approach 0

as the effective wave speed approaches infinity.

This can be shown by taking the limit as b approaches infinity in (25):

$$\begin{aligned} \lim_{b \rightarrow \infty} P_2 &= \lim_{b \rightarrow \infty} \frac{4 \int_0^1 \frac{dy}{(\sqrt{1-y^2})(\sqrt{1-e^2 y^2})}}{\omega} \\ &= \lim_{b \rightarrow \infty} \left(4 \int_0^1 \frac{dy}{(\sqrt{1-y^2})(\sqrt{1-e^2 y^2})} \right) \times \lim_{b \rightarrow \infty} \left(\frac{1}{\sqrt{a+bA^2}} \right) \end{aligned} \quad (32)$$

This requires the evaluation of $\lim_{b \rightarrow \infty} e$.

From (29):

$$e^2 = \frac{\frac{b}{2} A^2}{a + bA^2}$$

$-\sqrt{\frac{\frac{b}{2} A^2}{a + bA^2}}$ is omitted as $e \in (0, 1) \Rightarrow e > 0$.

$$e = \sqrt{\frac{\frac{b}{2} A^2}{a + bA^2}}$$

Taking limit as b approaches infinity:

$$\begin{aligned}\lim_{b \rightarrow \infty} e &= \lim_{b \rightarrow \infty} \sqrt{\frac{\frac{b}{2}A^2}{a + bA^2}} \\ &= \lim_{b \rightarrow \infty} \sqrt{\frac{\frac{1}{2}}{\frac{a}{A^2b} + 1}}\end{aligned}$$

This requires the evaluation of $\lim_{b \rightarrow \infty} A^2$.

From (27):

$$\frac{a}{2}A^2 + \frac{b}{4}A^4 = \frac{1}{2}T'(0)^2 + \frac{a}{2}T(0)^2 + \frac{b}{4}T(0)^4$$

Using quadratic formula yields:

$$\begin{aligned}A^2 &= \frac{-a + \sqrt{(a + bT(0)^2)^2 + 2bT'(0)^2}}{b} \\ \lim_{b \rightarrow \infty} A^2 &= \lim_{b \rightarrow \infty} \frac{-a + \sqrt{(a + bT(0)^2)^2 + 2bT'(0)^2}}{b} \\ &= \lim_{b \rightarrow \infty} \frac{-a + b\sqrt{\left(\frac{a}{b} + T(0)^2\right)^2 + \frac{2T'(0)^2}{b}}}{b} \\ &= \lim_{b \rightarrow \infty} \frac{-\frac{a}{b} + \sqrt{\left(\frac{a}{b} + T(0)^2\right)^2 + \frac{2T'(0)^2}{b}}}{1} \\ &= \frac{-0 + \sqrt{(0 + T(0)^2)^2 + 0}}{1} \\ &= T(0)^2\end{aligned}$$

Therefore $\lim_{b \rightarrow \infty} A^2 = T(0)^2$ is a constant, thus $\lim_{b \rightarrow \infty} \frac{a}{A^2 b}$ would approach 0.

$$\begin{aligned}\lim_{b \rightarrow \infty} e &= \lim_{b \rightarrow \infty} \sqrt{\frac{\frac{1}{2}}{\frac{a}{A^2 b} + 1}} \\ &= \sqrt{\frac{\frac{1}{2}}{0 + 1}} \\ &= \frac{1}{\sqrt{2}} \\ &\approx 0.707\end{aligned}$$

Use this result and continue from (32):

$$\begin{aligned}\lim_{b \rightarrow \infty} P_2 &= \lim_{b \rightarrow \infty} \left[\left(4 \int_0^1 \frac{dy}{(\sqrt{1-y^2})(\sqrt{1-e^2 y^2})} \right) \times \lim_{b \rightarrow \infty} \left(\frac{1}{\sqrt{a+bA^2}} \right) \right] \\ &= 4 \int_0^1 \frac{dy}{(\sqrt{1-y^2})(\sqrt{1-(0.707)^2 y^2})} \times 0 \\ &= 0\end{aligned}$$

This proves that theoretically, as the nonlinear term b becomes infinitely large, the wave would oscillate infinitely fast leading the period to approach 0.

2. When b approaches 0

Conversely, if $b \rightarrow 0$ then P_2 approaches P_1 as there is no longer any nonlinearity

as the PDE becomes identical to the linear wave:

$$\frac{\partial^2 y}{\partial t^2} - V^2(1+0) \left(\frac{\partial^2 y}{\partial x^2} \right) = 0$$

This can also be shown by taking the limit:

$$\lim_{b \rightarrow 0} P_2 = \lim_{b \rightarrow 0} \frac{4 \left(\int_0^1 \frac{dy}{(\sqrt{1-y^2})(\sqrt{1-e^2 y^2})} - \varphi \right)}{\sqrt{a + bA^2}} \quad (33)$$

This requires the evaluation of $\lim_{b \rightarrow 0} A^2$.

$$\lim_{b \rightarrow 0} A^2 = \lim_{b \rightarrow 0} \frac{-a + \sqrt{(a + bT(0)^2)^2 + 2bT'(0)^2}}{b}$$

Applying L'Hôpital's rule yields:

$$\begin{aligned} \lim_{b \rightarrow 0} \frac{-a + \sqrt{(a + b[T(0)]^2)^2 + 2b[T'(0)]^2}}{b} &= \lim_{b \rightarrow 0} \frac{2(a + bT(0))[T(0)]^2 + 2[T'(0)]^2}{2\sqrt{(a + bT(0)^2)^2 + 2bT'(0)^2}} \\ &= \frac{2(a)[T(0)]^2 + 2[T'(0)]^2}{2a} \\ &= [T(0)]^2 + \frac{[T'(0)]^2}{a} \end{aligned}$$

Thus, e approaches 0 as b approaches 0 because:

$$\begin{aligned} \lim_{b \rightarrow 0} e &= \lim_{b \rightarrow 0} \sqrt{\frac{\frac{b}{2} A^2}{a + bA^2}} \\ &= \sqrt{\frac{\frac{0}{2} \left([T(0)]^2 + \frac{[T'(0)]^2}{a} \right)}{a + 0 \left([T(0)]^2 + \frac{[T'(0)]^2}{a} \right)}} \\ &= 0 \end{aligned}$$

Continuing from (33):

$$\begin{aligned}
\lim_{b \rightarrow 0} P_2 &= \lim_{b \rightarrow 0} \frac{4 \left(\int_0^1 \frac{dy}{(\sqrt{1-y^2})(\sqrt{1-e^2 y^2})} - \varphi \right)}{\sqrt{a + bA^2}} \\
&= \frac{4 \left(\int_0^1 \frac{dy}{(\sqrt{1-y^2})(\sqrt{1-0^2 y^2})} - \varphi \right)}{\sqrt{a + (0) \left([T(0)]^2 + \frac{[T'(0)]^2}{a} \right)}} \\
&= \frac{4 \left(\int_0^1 \frac{dy}{(\sqrt{1-y^2})} - \varphi \right)}{\sqrt{a}}
\end{aligned}$$

Substituting $a = 2.34 \times 10^6$ and $\varphi = 0$ yields:

$$\begin{aligned}
\lim_{b \rightarrow 0} P_2 &= \frac{4 \left(\frac{\pi}{2} \right)}{\sqrt{2.34 \times 10^6}} \\
\lim_{b \rightarrow 0} P_2 &= 4.11 \times 10^{-3} = P_1
\end{aligned}$$

3.4 Difference 2: instantaneous frequency

The nonlinear wave has a non-constant frequency whilst the linear wave has a constant frequency.

As this behavior is not easily observable with the current parameters, evidence of this difference is shown by modifying the nonlinear wave such that its eccentricity of is close to 1 (0.99 was chosen). Modifying both the linear and nonlinear wave's period to $P = 2$ using (31) causing their average frequency to become equal.

Applying these modifications, the nonlinear wave becomes:

$$y(x, t) = \cos\left(\frac{4K(0.99)}{2}t, 0.99\right) \sin(4.85x) \approx \cos(6.71t, 0.99) \sin(4.85x)$$

The linear wave becomes:

$$y(x, t) = \cos(3.14t) \sin(4.85x)$$

while the linear wave is unchanged.

The following is an animation of the linear wave and nonlinear wave after modification:

[https://drive.google.com/file/d/1R8UNSOXahVFF2A73Cyo2WRxVGIZR2lzs/view?](https://drive.google.com/file/d/1R8UNSOXahVFF2A73Cyo2WRxVGIZR2lzs/view?usp=drive_link)

[usp=drive_link](https://drive.google.com/file/d/1R8UNSOXahVFF2A73Cyo2WRxVGIZR2lzs/view?usp=drive_link)

(“Give python code that graphs $\cos(6.71t, 0.99)$ and $\cos(3.14t)$ for one period)

(Appendix C)

Notice that the nonlinear wave moves faster than the linear wave when close to maximum amplitude and moves slower than the linear wave when close to equilibrium position (when the wave looks like a horizontal line).

The following shows the temporal part of the linear wave and nonlinear wave after modification:

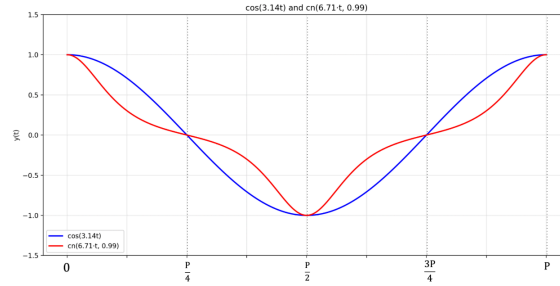


Diagram 14: comparing temporal part of solution for linear and nonlinear wave

(“Give python code that graphs $\text{cn}(6.71t, 0.99)$ and $\cos(3.14t)$ for one period”)

(Appendix D)

Notice that the slope of $\text{cn}(6.71t, 0.99)$ (that is, the speed of the nonlinear wave) is

higher than $\cos(3.14t)$ (that is, the speed of the linear wave) for t close to

$t = \frac{(n-1)P}{2}$ for $n \in \mathbb{Z}^+$ and that the slope of $\text{cn}(6.71t, 0.99)$ is lower than

$\cos(3.14t)$ (that is, the speed of the linear wave) for t close to

$t = \frac{(2n-1)P}{4}$ for $n \in \mathbb{Z}^+$.

3.4.1 Qualitative explanation

Since the linear wave has a constant instantaneous frequency which is the same as the

nonlinear wave’s average frequency, the fact that nonlinear wave moves faster than

the linear wave when close to maximum amplitude (that is, when t is close to

$t = \frac{(n-1)P}{2}$ for $n \in \mathbb{Z}^+$) suggests that, at those times, the nonlinear wave’s frequency

is higher than its average frequency.

Conversely, the fact that nonlinear wave moves slower than the linear wave when close to equilibrium (that is, when t is close to $t = \frac{(2n-1)P}{4}$ for $n \in \mathbb{Z}^+$) suggests that, at those times, its frequency is higher than its average frequency.

This shows that, unlike the linear wave, the instantaneous frequency of the nonlinear wave varies over time, becoming higher than its average frequency at t close to $t = \frac{(n-1)P}{2}$ and lower than its average frequency at t close to $t = \frac{(2n-1)P}{4}$.

3.4.2 Quantitative explanation

The behaviour can also be explained without modification by explicitly finding the instantaneous frequency of $y(x, t) = \cos(1530t) \sin(4.85x)$ and $y(x, t) = \text{cn}(62200t, 0.701) \sin(4.85x)$.

This is achieved by considering that the instantaneous frequency of any oscillatory signal $s(t)$ is given by the following relation:

$$f_{inst}(t) = \frac{1}{2\pi} \frac{d\Phi(t)}{dt}$$

Where $s(t) = A \cos(\Phi(t))$ (Baoshash).

For the temporal part of the linear wave:

$$s(t) = A \cos(\omega t)$$

Therefore:

$$\Phi(t) = \omega t$$

$$f_{inst}(t) = \frac{\omega}{2\pi}$$

Substituting $\omega = 1530$:

$$f_{inst}(t) \approx 244$$

For the temporal part of the nonlinear wave:

$$s(t) = A \text{cn}(\omega t + \varphi, e)$$

This implies:

$$A \cos(\Phi(t)) = A \text{cn}(\omega t + \varphi, e)$$

$$\Phi(t) = \arccos(\text{cn}(\omega t + \varphi, e))$$

Therefore, in the nonlinear case:

$$\begin{aligned} f_{inst}(t) &= \frac{1}{2\pi} \frac{d\Phi(t)}{dt} \\ &= \frac{1}{2\pi} \frac{d}{dt} (\arccos(\text{cn}(\omega t + \varphi, e))) \end{aligned}$$

Applying chain rule

$$f_{inst}(t) = -\frac{1}{2\pi} \frac{1}{\sqrt{1 - \text{cn}^2(\omega t + \varphi, e)}} \frac{d}{dt} (\text{cn}(\omega t + \varphi, e)) \quad (34)$$

To evaluate $\frac{d}{dt} (\text{cn}(\omega t + \varphi, e))$, (22) is differentiated with respect to time:

$$\begin{aligned} \frac{d}{dt} (\text{sn}^2(u, e) + \text{cn}^2(u, e)) &= \frac{d}{dt} (1) \\ 2\text{cn}(\omega t + \varphi, e) \left(\frac{d}{dt} (\text{cn}(\omega t + \varphi, e)) \right) + 2\text{sn}(\omega t + \varphi, e) \left(\frac{d}{dt} (\text{sn}(\omega t + \varphi, e)) \right) &= 0 \quad (35) \end{aligned}$$

From (24):

$$\begin{aligned}\frac{d}{dt}(\text{sn}(u, e)) &= \text{cn}(u, e)\text{dn}(u, e) \\ \Rightarrow \frac{d}{dt}(\text{sn}(\omega t + \varphi, e)) &= \omega \text{cn}(\omega t + \varphi, e)\text{dn}(\omega t + \varphi, e)\end{aligned}$$

Substituting this into (35) yields:

$$\begin{aligned}2\text{cn}(\omega t + \varphi, e) \left(\frac{d}{dt}(\text{cn}(\omega t + \varphi, e)) \right) + 2\omega \text{sn}(\omega t + \varphi, e)\text{cn}(\omega t + \varphi, e)\text{dn}(\omega t + \varphi, e) &= 0 \\ \frac{d}{dt}(\text{cn}(\omega t + \varphi, e)) + \omega \text{sn}(\omega t + \varphi, e)(\text{dn}(\omega t + \varphi, e)) &= 0 \\ \frac{d}{dt}(\text{cn}(\omega t + \varphi, e)) &= -\omega \text{sn}(\omega t + \varphi, e)(\text{dn}(\omega t + \varphi, e))\end{aligned}$$

Continuing from (34):

$$\begin{aligned}f_{inst}(t) &= -\frac{1}{2\pi} \frac{1}{\sqrt{1 - \text{cn}^2(\omega t + \varphi, e)}} \frac{d}{dt}(\text{cn}(\omega t + \varphi, e)) \\ &= \frac{1}{2\pi} \frac{\omega}{\text{sn}(\omega t + \varphi, e)} [(\text{sn}(\omega t + \varphi, e))(\text{dn}(\omega t + \varphi, e))] \\ &= \frac{\omega \text{dn}(\omega t + \varphi, e)}{2\pi}\end{aligned}$$

Substituting $\varphi = 0$, $\omega = 62200$, $e = 0.701$ yields:

$$f_{inst}(t) = \frac{62200 \text{dn}(62200t, 0.701)}{2\pi} \approx 9900 \text{dn}(62200t, 0.701)$$

The following graph shows $f_{inst}(t)$ for both the linear and nonlinear wave.

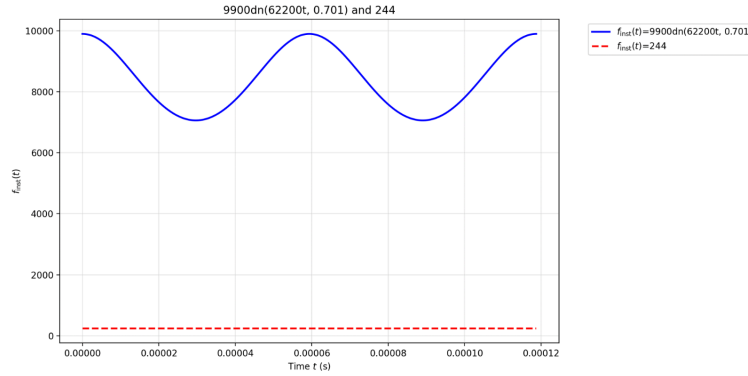


Diagram 15: instantaneous frequency of the nonlinear and linear wave

(“Give python code that graphs 9900dn(62200t, 0.701) and 244 from t=0 to t=0.000119”) (Appendix E)

While the instantaneous frequency of $\cos(3.14t)$ remains constant, the instantaneous frequency of the $\text{cn}(62200t, 0.701)$ varies in a wave like pattern.

As predicted in the qualitative explanation, the instantaneous frequency of $\text{cn}(62200t, 0.701)$ decreases to a minimum whenever t approaches

$$\frac{(2n-1)P_2}{2} \text{ for } n \in \mathbb{Z}^+ \text{ and increases to a maximum as } t \text{ approaches}$$

$$\frac{(n-1)P_2}{2} \text{ for } n \in \mathbb{Z}^+.$$

3.5 Difference 3: Range of possible initial conditions

As discussed previously, a limitation to the analytical solution to the nonlinear case is that it is only applicable to initial conditions which excite one mode.

However, for the analytical solution to the linear wave there is no such restriction.

Therefore, arbitrary functions can be set as the initial conditions for the linear wave,

this complicates the temporal and spatial part of the linear wave.

For example, setting $f(x) = 1$ and $g(x) = -1$ yields:

$$B_n \sin\left(\frac{n\pi}{L}x\right) \neq 1, \text{ for } n \in \mathbb{N}$$

$$\frac{\partial y}{\partial t} = -\frac{\pi n A_n V}{L} \sin\left(\frac{n\pi}{L}x\right) \neq -1, \text{ for } n \in \mathbb{N}$$

There no longer exists $\sin\left(\frac{n\pi}{L}x\right)$ which matches with the initial conditions

Therefore, an infinite sum of sines is required to construct $f(x)$ and $g(x)$.

The coefficients of those sine functions are given by the orthogonality relations as

derived previously.

Explicitly,

$$y(x, t) = \sum_{n=1}^{\infty} (A_n \sin(1530nt) + B_n \cos(1530nt)) (\sin(4.85nx))$$

Where:

$$B_n = 3.09 \int_0^{0.648} \sin(4.85nx) dx$$

$$A_n = -\frac{2.01 \times 10^{-3}}{n} \int_0^{0.648} \sin(4.85nx) dx$$

This is a snapshot of the resultant propagating wave

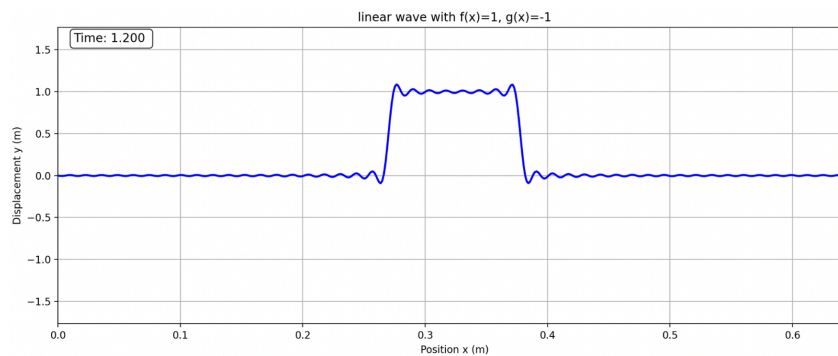


Diagram 16: linear wave with arbitrary initial conditions

(“Give python code that animates the analytical solution in the following form:

$$y(x, t) = \sum_{n=1}^{\infty} \left(A_n \sin\left(\frac{n\pi}{L} V t\right) + B_n \cos\left(\frac{n\pi}{L} V t\right) \right) \left(\sin\left(\frac{n\pi}{L} x\right) \right) \text{ Where } B_n = 3.09 \int_0^{0.648} \sin(4.85nx) dx \text{ and } A_n = -\frac{2.01 \times 10^{-3}}{n} \int_0^{0.648} \sin(4.85nx) dx ”)$$

(Appendix F)

Because the temporal part consists of an infinite sum of sines and cosines, each consisting of its own frequency, the resultant wave’s frequency is no longer a single number but a combination of the frequency of all constituent modes.

Unlike monomodal oscillations, the spatial part of the solution consists of an infinite sum of sines. This causes the shape of the wave to no longer be a single sinusoidal arch, instead taking a more complex wave form.

4. Conclusion

This essay compares the analytical solution of a propagating wave on a string fixed on both ends.

The analytical solution without the consideration of stretching is:

$$y(x, t) = \sum_{n=1}^{\infty} \left(A_n \sin\left(\frac{n\pi}{L} Vt\right) + B_n \cos\left(\frac{n\pi}{L} Vt\right) \right) \left(\sin\left(\frac{n\pi}{L} x\right) \right)$$

The analytical solution with the consideration of stretching is:

$$y(x, t) = A \operatorname{cn}(\pm \omega t + \Phi, e) \left(\sin\left(\frac{n\pi}{L} x\right) \right)$$

In both cases, similar techniques are used to separate the PDE into a temporal component and spatial component. Because both spatial components are identical, the waveform of both analytical solutions is similar. However, differences in the temporal part of the solution cause the 2 solutions to differ in terms of period and instantaneous frequency. Additionally, the range of possible initial conditions is much smaller for the 2nd case compared to the 1st case.

The significance of the assumptions in this research varies: although musical strings are generally assumed to be a uniform line of material well described under the assumption of low amplitudes (Fletcher and Rossing), frictional losses become dominant in guitar strings during small amplitude oscillations (Rossing). As an

extension, solutions to the 2nd case for arbitrary initial conditions can be explored by introducing numerical methods such as perturbation method or finite difference method. Furthermore, the aforementioned numerical methods could be used to compare with the analytical solutions.

5. Appendix

A Python code for animation shown in Diagram 12:

```
import numpy as np
import matplotlib.pyplot as plt
from matplotlib.animation import FuncAnimation
from scipy.special import ellipj
import warnings
warnings.filterwarnings('ignore')

# Set up the figure and axes
fig, ax = plt.subplots(figsize=(12, 6))

# Define the x domain
x = np.linspace(0, 0.648, 500)

# Define the time range and create time array
t_start, t_end = 0, 0.005
t_values = np.linspace(t_start, t_end, 200)

# Initialize the plots
line1, = ax.plot(x, np.zeros_like(x), 'b-', linewidth=2, label='cn(62200t, 0.701)sin(4.85x)')
line2, = ax.plot(x, np.zeros_like(x), 'r-', linewidth=2, label='cos(1530t)sin(4.85x)')

# Set up the axes
ax.set_xlim(0, 0.648)
ax.set_ylim(-1.2, 1.2)
ax.set_xlabel('x')
ax.set_ylabel('y(x,t)')
ax.grid(True, alpha=0.3)
ax.legend(loc='upper right')
```

```

# Create a timer box in the top-left corner
timer_box = ax.text(0.02, 0.98, 'Time: 0.00000 s', transform=ax.transAxes,
                    fontsize=12, verticalalignment='top',
                    bbox=dict(boxstyle='round', facecolor='white', alpha=0.8,
                              edgecolor='black'))

# Function 1: Using Jacobi elliptic cosine (cn)
def function1(x, t):
    # For scipy, ellipj(u, m) where m = k^2
    # Here k = 0.701, so m = 0.701^2
    m = 0.701**2
    u = 62200 * t
    # ellipj returns (sn, cn, dn, ph)
    cn_val = ellipj(u, m)[1]
    return cn_val * np.sin(4.85 * x)

# Function 2: Absolute value of cosine times sine
def function2(x, t):
    return np.cos(1530 * t) * np.sin(4.85 * x)

# Animation update function
def update(frame):
    t = t_values[frame]

    # Update both functions
    y1 = function1(x, t)
    y2 = function2(x, t)

    line1.set_ydata(y1)
    line2.set_ydata(y2)

    # Update timer box with current time
    timer_box.set_text(f'Time: {t:.5f} s')

    return line1, line2, timer_box

```

```
animation = FuncAnimation(fig, update, frames=len(t_values),
                          interval=50, blit=False, repeat=True)
```

```
ax.set_title(f'cos(1530t)sin(4.85x) and cn(62200t, 0.701)sin(4.85x)')
```

```
plt.tight_layout()
plt.show()
```

B Python code for Diagram 13

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.special import ellipj

# Parameters
eccentricity = 0.701
frequency = 62200
t_end = 1.2e-4

# Create time array
t = np.linspace(0, t_end, 1000)

# Calculate the arguments for the elliptic functions
u = frequency * t
m = eccentricity**2 # scipy uses parameter m = k^2

# Calculate sn and cn values
sn_values, cn_values, dn_values, ph = ellipj(u, m)

plt.figure(figsize=(10, 6))
plt.plot(t, sn_values, 'b-', linewidth=2, label=f'sn(62200t, {eccentricity})')
plt.plot(t, cn_values, 'r-', linewidth=2, label=f'cn(62200t, {eccentricity})')
plt.xlabel('Time (s)')
plt.ylabel('Function Value')
plt.title(f'cn(62200t, 0.701) and sn(62200t, 0.701)')
plt.grid(True, alpha=0.3)
```

```
plt.legend()
plt.tight_layout()
plt.show()
```

C Python code that animation linear wave and nonlinear wave after modification

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.special import ellipk, ellipj

# Parameters
m = 0.491
A = 9900
f = 62200
horizontal_value = 244

# Calculate period of dn function
K = ellipk(m) # Complete elliptic integral of the first kind
period_u = 2 * K # Period in the u domain
period_t = period_u / f # Period in the t domain

# Create time array for two periods
t_two_periods = np.linspace(0, 2 * period_t, 1000)

# Compute dn function
u_vals = f * t_two_periods
# ellipj(u, m) returns (sn, cn, dn, ph)
_, _, dn_vals, _ = ellipj(u_vals, m)
f_dn_vals = A * dn_vals

# Create the horizontal line array
f_horizontal = np.full_like(t_two_periods, horizontal_value)

# Plot
plt.figure(figsize=(12, 6))
plt.plot(t_two_periods, f_dn_vals, 'b-', label=r'$f_{\mathrm{inst}}(t) = 9900\mathrm{dn}(62200t, 0.701)$', linewidth=2)
```

```
plt.plot(t_two_periods, f_horizontal, 'r--', label=r'$f_{\mathrm{inst}}(t) \text{ } \$=244' ,
linewidth=2)
```

```
plt.xlabel('Time $t$ (s)')
plt.ylabel('$f_{\mathrm{inst}}(t)$')
plt.title('9900dn(62200t, 0.701) and 244')
plt.grid(True, alpha=0.3)
```

```
# Move legend outside the graphing area
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
```

```
plt.tight_layout()
plt.show()
```

D Python code for Diagram 14:

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.special import ellipj
from scipy.special import ellipk

def cn(u, m):
    """Jacobi elliptic function cn(u, m)"""
    result = ellipj(u, m)
    return result[1]

# Parameters
m = 0.99 # parameter for cn function
K = ellipk(m) # Complete elliptic integral of the first kind

# cn(2K·t, m) has period 1 (since cn(2K·(t+1), m) = cn(2K·t + 4K, m) = cn(2K·t, m))
# cos(πt) has period 2
# To compare them properly, let's plot over a range that shows their periods clearly

# Create time array (showing multiple periods)
t = np.linspace(0, 2, 2000) # 4 units to show multiple periods
```

```

# Calculate functions
cos_pi_t = np.cos(np.pi * t)
cn_2K_t = cn(2 * K * t, m)

# Create the plot
plt.figure(figsize=(12, 6))

plt.plot(t, cos_pi_t, 'b-', linewidth=2, label='cos(3.14t) ')
plt.plot(t, cn_2K_t, 'r-', linewidth=2, label=f'cn(6.71·t, {m}) ')

plt.xlabel('Time (t)')
plt.ylabel('y(t)')
plt.title('cos(3.14t) and cn(6.71·t, 0.99)')
plt.legend(loc='lower left')
plt.grid(True, alpha=0.3)
plt.ylim(-1.5, 1.5)

plt.tight_layout()
plt.show()

```

E Python code for Diagram 15:

```

import numpy as np
import matplotlib.pyplot as plt
from scipy.special import ellipk, ellipj

# Parameters
m = 0.491
A = 9900
f = 62200
horizontal_value = 244

# Calculate period of dn function
K = ellipk(m) # Complete elliptic integral of the first kind
period_u = 2 * K # Period in the u domain
period_t = period_u / f # Period in the t domain

# Create time array for two periods

```



```

t_two_periods = np.linspace(0, 2 * period_t, 1000)

# Compute dn function
u_vals = f * t_two_periods
# ellipj(u, m) returns (sn, cn, dn, ph)
_, _, dn_vals, _ = ellipj(u_vals, m)
f_dn_vals = A * dn_vals

# Create the horizontal line array
f_horizontal = np.full_like(t_two_periods, horizontal_value)

# Plot
plt.figure(figsize=(12, 6))
plt.plot(t_two_periods, f_dn_vals, 'b-', label=r'$f_{\mathrm{inst}}(t) = 9900\mathrm{dn}(62200t, 0.701)$', linewidth=2)
plt.plot(t_two_periods, f_horizontal, 'r--', label=r'$f_{\mathrm{inst}}(t) = 244$', linewidth=2)

plt.xlabel('Time $t$ (s)')
plt.ylabel('$f_{\mathrm{inst}}(t)$')
plt.title('9900dn(62200t, 0.701) and 244')
plt.grid(True, alpha=0.3)

# Move legend outside the graphing area
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')

plt.tight_layout()
plt.show()

```

F Python code for animation shown in Diagram 16:

```

import numpy as np
import matplotlib.pyplot as plt
from matplotlib.animation import FuncAnimation
from scipy.integrate import quad

# Parameters

```

```

L = 0.648 # Length of the string
N_terms = 100 # Number of Fourier coefficients
t_max = 2.0 # Maximum time in the physical system
x_points = 500 # Increased spatial resolution

# Calculate Fourier coefficients B_n and A_n more carefully
B_n = np.zeros(N_terms)
A_n = np.zeros(N_terms)

print("Calculating Fourier coefficients...")
for n in range(1, N_terms + 1):
    # Calculate the integral  $\int \sin(4.85nx)dx$  from 0 to L
    # Analytical solution:  $\int \sin(kx)dx = -\cos(kx)/k$ 
    k = 4.85 * n
    integral = (-np.cos(k * L) + np.cos(0)) / k
    # integral = (1 - np.cos(k * L)) / k

    B_n[n-1] = (2 / L) * integral
    A_n[n-1] = (-2.01e-3 / n) * integral

# Print first few coefficients for debugging
if n <= 5:
    print(f'n={n}: B_n = {B_n[n-1]:.6f}, A_n = {A_n[n-1]:.6f}')

# Check coefficient magnitudes
print(f'\nMax |B_n|: {np.max(np.abs(B_n)):.6f}')
print(f'Max |A_n|: {np.max(np.abs(A_n)):.6f}')

# Create spatial grid
x = np.linspace(0, L, x_points)

# Test the function at t=0 to see initial shape
def string_displacement(t):
    y = np.zeros_like(x)
    for n in range(1, N_terms + 1):
        term = (A_n[n-1] * np.sin(1530 * n * t) +
                B_n[n-1] * np.cos(1530 * n * t)) * np.sin(4.85 * n * x)
        y += term

```

```

    return y

# Check initial displacement
y_initial = string_displacement(0)
print(f"\nInitial displacement range: [{np.min(y_initial):.6f}, {np.max(y_initial):.6f}]\n")

# Set up the figure and axis with appropriate limits
fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(12, 10))

# Main animation plot
ax1.set_xlim(0, L)
# Auto-scale y-axis based on initial displacement
y_range = np.max(np.abs(y_initial)) * 1.5 if np.max(np.abs(y_initial)) > 0 else 0.01
ax1.set_ylim(-y_range, y_range)
ax1.set_xlabel('Position x (m)')
ax1.set_ylabel('Displacement y (m)')
ax1.set_title('linear wave with f(x)=1, g(x)=-1')
ax1.grid(True)

# Coefficient plot
n_values = np.arange(1, N_terms + 1)
ax2.semilogy(n_values, np.abs(B_n), 'ro-', markersize=2, label='|B_n|', alpha=0.7)
ax2.semilogy(n_values, np.abs(A_n), 'bo-', markersize=2, label='|A_n|', alpha=0.7)
ax2.set_xlabel('Fourier coefficient index n')
ax2.set_ylabel('Magnitude (log scale)')
ax2.set_title('Fourier Coefficient Magnitudes')
ax2.legend()
ax2.grid(True)

# Initialize the line
line, = ax1.plot([], [], 'b-', linewidth=2)

# Add time display
time_text = ax1.text(0.02, 0.95, "", transform=ax1.transAxes, fontsize=12,
                    bbox=dict(boxstyle="round,pad=0.3", facecolor="white", alpha=0.8))

# Animation function
def animate(frame):

```

```

# Convert frame to physical time (10x slower: 1s animation = 10s real)
t_physical = frame * (t_max / 200) # 200 frames total

# Calculate string displacement
y = string_displacement(t_physical)

# Update the line
line.set_data(x, y)

# Update time display
time_text.set_text(f'Time: {t_physical:.3f} ')

return line, time_text

# Create animation
print("\nCreating animation...")
animation = FuncAnimation(fig, animate, frames=200, interval=50, blit=True)

plt.tight_layout()
plt.show()

# Additional diagnostic plot
plt.figure(figsize=(10, 6))
for t in [0, 0.1, 0.2, 0.5]:
    y = string_displacement(t)
    plt.plot(x, y, label=f't = {t}s')
plt.xlabel('Position x (m)')
plt.ylabel('Displacement y (m)')
plt.title('linear wave with f(x)=1, g(x)=-1')
plt.legend()
plt.grid(True)
plt.show()

```

7. Works Cited

"All About Wave Equation." Unacademy, unacademy.com/content/neet-ug/study-material/physics/all-about-wave-equation/.

"An Introduction to Ordinary Differential Equations." Math Insight, mathinsight.org/ordinary_differential_equation_introduction.

Coulson, C. A. Waves: A Mathematical Account of the Common Types of Wave Motion. 7th ed., Longman, 1977. Universidad de la República, Uruguay, eva.fcien.udelar.edu.uy/pluginfile.php/143665/mod_resource/content/1/Coulson.-.Waves.%281977%29%28T%29%28237s%29.pdf.

"Ellipse." BYJU'S, byjus.com/maths/ellipse/.

Fitzpatrick, Richard. "Standing Waves." The University of Texas at Austin, farside.ph.utexas.edu/teaching/315/Waveshtml/node68.html.

“Give python code that animate the graphs of $\cos(6.71t, 0.99)\sin(4.85x)$ and $\cos(3.14t)\sin(4.85x)$ from $t=0$ to $t=0.005$ ” prompt. Deepseek, model V3.1, High-Flyer, 24 June 2025, <https://www.deepseek.com/>.

“Give python code that animate the graphs of $\cos(1530t)\sin(4.85x)$ and $\cos(62200t, 0.701)\sin(4.85x)$ from $t=0$ to $t=0.005$ ” prompt. Deepseek, model V3.1, High-Flyer, 24 June 2025, <https://www.deepseek.com/>.

“Give python code that animates the analytical solution in the following form:

$$y(x, t) = \sum_{n=1}^{\infty} \left(A_n \sin\left(\frac{n\pi}{L} V t\right) + B_n \cos\left(\frac{n\pi}{L} V t\right) \right) \left(\sin\left(\frac{n\pi}{L} x\right) \right) \text{ Where } B_n = \frac{2}{L} \int_0^{0.648} \sin(4.85nx) dx \text{ and } A_n = -\frac{2.01 \times 10^{-3}}{n} \int_0^{0.648} \sin(4.85nx) dx ” prompt.$$

Deepseek, model V3.1, High-Flyer, 24 June 2025, <https://www.deepseek.com/>.

“Give python code that graphs $9900\sin(62200t, 0.701)$ and 244 from $t=0$ to $t=0.000119$ ” prompt. Deepseek, model V3.1, High-Flyer, 24 June 2025, <https://www.deepseek.com/>.

“Give python code that graphs $\cos(6.71t, 0.99)$ and $\cos(3.14t)$ for one period” prompt. Deepseek, model V3.1, High-Flyer, 24 June 2025, <https://www.deepseek.com/>.

“Give python code that graphs $\text{cn}(62200t, 0.701)$ and $\text{sn}(62200t, 0.701)$ for one period ” prompt. Deepseek, model V3.1, High-Flyer, 24 June 2025, <https://www.deepseek.com/>.

Herman, Russell L. Partial Differential Equations: A Comprehensive Introduction. 2017. University of North Carolina Wilmington, people.uncw.edu/hermanr/pdel/pdebook/PDEMain.pdf. Accessed 13 June 2025.

Higa, Meg. "What Is the Principle of Superposition? (with Picture)." All the Science, 21 July 2023, www.allthescience.org/what-is-the-principle-of-superposition.htm. Accessed 3 Aug. 2025.

Kovacic, Ivana, Livija Cveticanin, Miodrag Zukovic, and Zvonko Rakaric. “Jacobi Elliptic Functions: A Review of Nonlinear Oscillatory Application Problems.” Journal of Sound and Vibration, vol. 380, 13 Oct. 2016, pp. 1–36. Elsevier, <https://doi.org/10.1016/j.jsv.2016.05.051>.

"Modulus of Elasticity." Instron, Instron, 2024,

www.instron.com/en/resources/glossary/modulus-of-elasticity/. Accessed 1 July. 2025.

Morse, Philip M., and K. Uno Ingard. Theoretical Acoustics. Princeton UP, 1986.

Neumann, Christian, and Marie-Ann Guenther. "From Neumann to Dirichlet Boundary

Conditions for the 1D Wave Equation." WRAP: Warwick Research Archive Portal,

University of Warwick, 2022, [wrap.warwick.ac.uk/167742/7/WRAP-Neumann-](http://wrap.warwick.ac.uk/167742/7/WRAP-Neumann-Dirichlet-boundary-conditions-one-dimensional-wave-equation-22.pdf)

[Dirichlet-boundary-conditions-one-dimensional-wave-equation-22.pdf](http://wrap.warwick.ac.uk/167742/7/WRAP-Neumann-Dirichlet-boundary-conditions-one-dimensional-wave-equation-22.pdf). Accessed 15

Oct. 2023.

[pressbooks.online.ucf.edu/osuniversityphysics/chapter/16-6-standing-waves-and-](http://pressbooks.online.ucf.edu/osuniversityphysics/chapter/16-6-standing-waves-and-resonance/)

[resonance/](http://pressbooks.online.ucf.edu/osuniversityphysics/chapter/16-6-standing-waves-and-resonance/). Accessed 23 May. 2025.

Sommerfeld, Arnold. Lectures on Theoretical Physics: Optics. Translated by Otto

Laporte and Peter A. Moldauer, Academic Press, 1954. Internet Archive,

archive.org/details/dli.ernet.11707/page/n25/mode/2up.

“Tension Force - Definition, Explanation, Solved Examples, and FAQs.” BYJU'S, BYJU'S, 2024, byjus.com/physics/tension-force/.

"The Physics of Musical Instruments." University of Waterloo Computer Science Club, uploaded by Patrick Barfuss, 2005, csclub.uwaterloo.ca/~pbarfuss/The_Physics_of_Musical_Instruments.pdf.

The Science of String Instruments. Edited by Thomas D. Rossing, The Logos Foundation, 2010, logosfoundation.org/kursus/The%20Science%20of%20String%20Instruments.pdf.

"The Story of Acoustics." The Journal of the Acoustical Society of America, vol. 39, no. 4, 1966, pp. 629-44. JASA, doi: 10.1121/1.1907948.

University of Illinois PDF Lecture Notes Dutta, Sudesna. "Waves 2: The Wave Equation." *University of Illinois at Urbana-Champaign, Physics 406*,

courses.physics.illinois.edu/phys406/sp2017/Lecture_Notes/Waves/PDF_Files/Waves2.pdf.

"Wave Equation." Brilliant Math & Science Wiki, brilliant.org/wiki/wave-equation/.

Weisstein, Eric W. "Elliptic Integral of the First Kind." MathWorld--A Wolfram Web Resource, Wolfram Research, Inc.,
mathworld.wolfram.com/EllipticIntegraloftheFirstKind.html.

Whittaker, E. T., and G. N. Watson. A Course of Modern Analysis. Cambridge University Press, 2021.